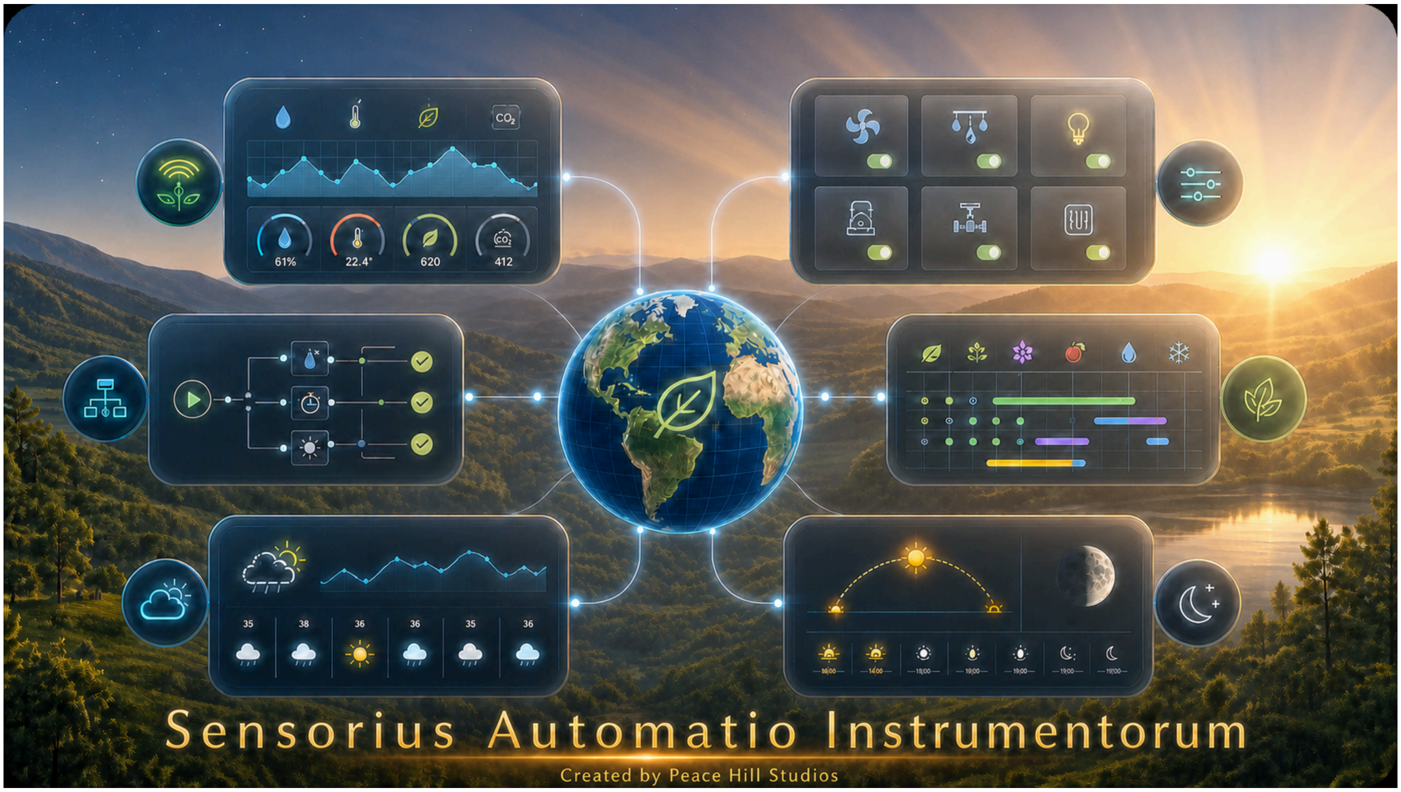


Sensorius User Guide

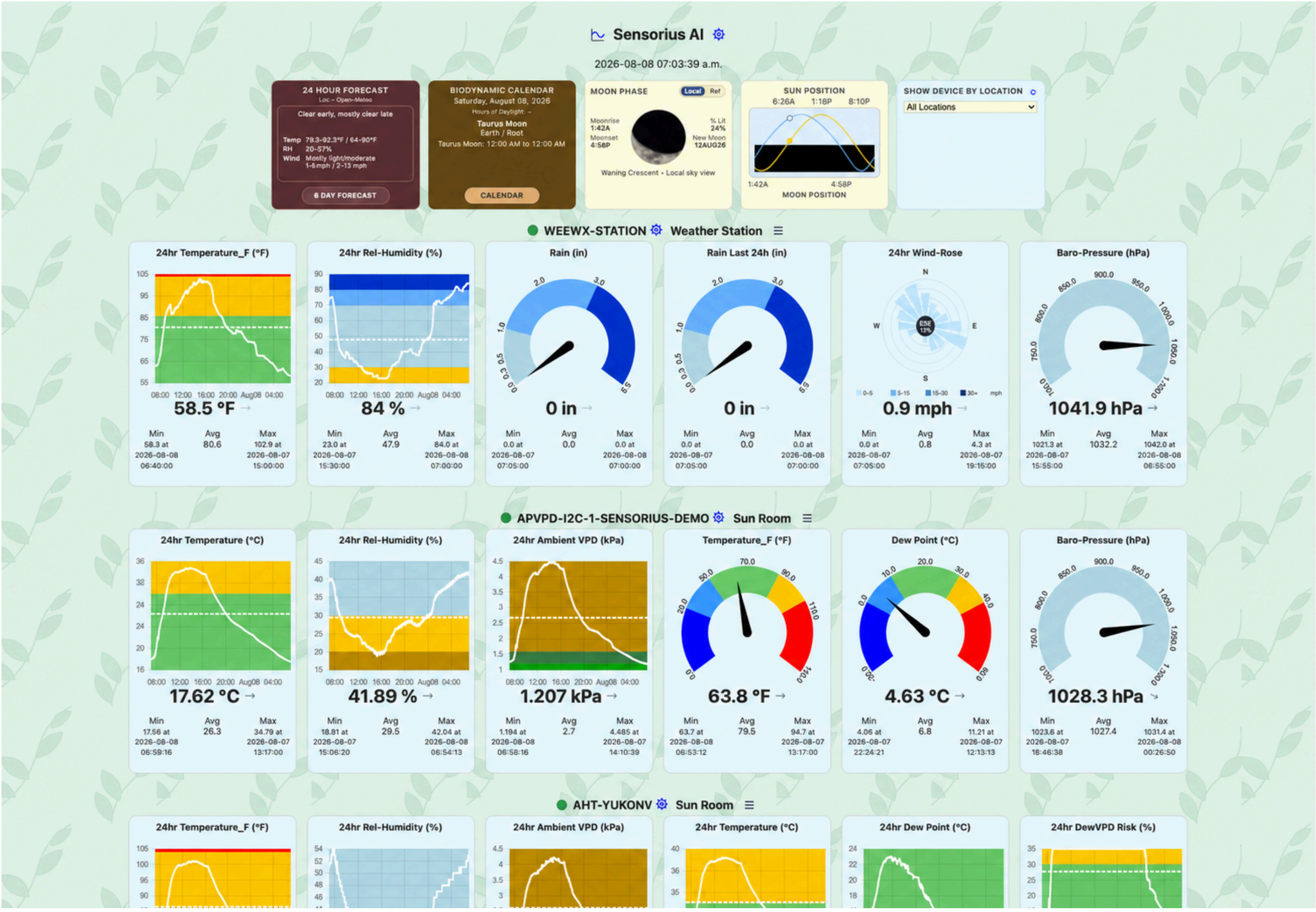
Sensorius Automatio Instrumentorum, also called Sensorius AI or Sensorius, is an environmental sensing and automation hub for gardens, greenhouses, grow rooms, small farms, and other places where environmental conditions matter. It gives you live readings, historical graphs, switch control, calibration tools, optional integrations with Home Assistant, WeeWX, and farmOS, and a fully integrated Biodynamic Calendar.

Sensorius can run as a Raspberry Pi hub with directly connected sensors and relays, as well as Wi-Fi Nodus sensors and switches that communicate through MQTT. It can also run on macOS, Windows, or Linux as a hub for Wi-Fi Nodus sensors and switches. In normal use, both kinds of devices appear together on the same dashboard.

This guide is written for people who want to use the system without necessarily knowing all of its back-end functionality. You do not need to understand MQTT, SQLite, or Python to use the app, but the guide explains where information comes from so you can make good decisions when something looks wrong.

Dashboard Overview

The dashboard is the main operating view. It is where you check current conditions, see switch state, open settings, and review quick trends.



This example is based on a representative Sensorius dashboard and uses the neutral host identity `sensorius-demo`. Sensor names, locations, metrics, switch channels, forecast source, and live values will vary by installation.

The dashboard presents:

- Sensor cards grouped by device and location.
- Switch cards with live channel state and recent events.
- Sun, moon, biodynamic, and optional weather forecast cards when Astral location is available.
- Buttons for General Settings, the full-screen graph, sensor settings, switch settings, and calendar views.

The **24 Hour Forecast** card keeps its day-wide forecast summary, while its details include the maximum forecast precipitation chance, labeled as rain or snow from the forecast conditions. Its background reflects the current three-hour forecast window. Clear conditions use a sunny daytime sky or a starry night; cloudy conditions use gray daytime or nighttime skies; and precipitation adds rain streaks to the corresponding gray sky. This card appearance is independent of the theme selected for the full-screen Caelus display. Day artwork begins 30 minutes before the configured Astral sunrise to represent daybreak; night artwork begins 30 minutes after sunset to include dusk. The station timezone is used, and the boundary is checked once per minute.

Select **Lunar Calendar** on the dashboard's **Moon Phase** card to open the Caelus observer-local phase timeline: four previous phases, the live Moon and its current details, and four upcoming phases with dates. Select **Sun/Moon Position** to open the separate 29-day position graph. Close it with the **×** button in the card's top-right corner, by selecting the card, or by pressing Escape.

Dashboard data comes from the latest values in the live runtime cache and from the local database. If a device is offline, the latest stored reading may still be visible, but the online/offline state comes from live device status, MQTT heartbeat or availability messages, and recent packets.

Sensor Cards

Each sensor card shows the sensor ID or name, its location, and the selected metrics for that sensor. The 24-hour metric-card micrographs use consistent three-hour local AM/PM ticks, with a month/day marker at midnight. The 24-hour Min and Max timestamps also use local AM/PM time. The metric names come from the sensor's settings and the measurements that the device reports. Common metrics include temperature, relative humidity, absolute humidity, CO2, VPD, dew point, barometric pressure, soil moisture, soil temperature, soil pH, soil EC, soil nutrients, light, and PPFD.

Each metric tile can appear as:

- **Gauge**: shows the current value against a colored range.
- **6Hr Graph**: shows the last six hours inside the dashboard tile.
- **24Hr Graph**: shows the last 24 hours inside the dashboard tile.

Newly onboarded Nodus sensors use **24Hr Graph** for their per-metric display styles. A full day is long enough to show the daily diurnal patterns for heating, cooling, humidity, irrigation, and lighting cycle without opening the full-screen graph, so it is the most useful general-purpose trend view. **General Settings > Display Style** is the system-wide fallback where no per-metric style is supplied; its factory value is **Gauge**. Blank per-metric styles on directly connected local sensors also currently resolve to **Gauge**.

Click a metric tile to cycle **24Hr Graph → 6Hr Graph → Gauge → 24Hr Graph**. This is a convenient temporary view change for the current page. To make the choice persistent, open **Sensor Settings > Sensor Settings**, choose **Gauge**, **6Hr Graph**, or **24Hr Graph** for that metric slot, and click **Save**. **General Settings > Display Style** supplies the fallback when a sensor has no saved style.

The current value comes from the latest reading for that metric. A trend arrow beside it uses recent stored readings rather than comparing only the last two values. Most metrics use a least-squares fit across the latest 19 minutes. Barometric pressure uses up to three hours and identifies a shorter result as provisional. Existing database history populates these windows immediately after startup; Sensorius does not need to collect a new in-memory window. The arrow remains in a learning state until at least six samples cover five minutes.

Pointing up indicates a significant increase, slightly upward indicates a moderate increase, right indicates neutral, slightly downward indicates a moderate decrease, and down indicates a significant decrease. Intermediate rates produce intermediate angles, and the arrow moves smoothly when the dashboard refreshes. Hover over or focus the arrow to see its rate per hour, actual history window, and provisional state.

Trend strength is normalized to the metric's gauge span so unlike units share the same visual language. Pressure uses a narrower meteorological scale. The arrow communicates direction and relative strength, not whether the change is good or bad. The small history graph comes from `/graph-data`, which reads stored samples from the local database for that sensor and metric.

The solid line is the stored reading series. The dashed line is the arithmetic average of the visible graph window, repeated across the window as a reference line; it is not another sensor reading or an automation set point. The Min, Avg, and Max values below each tile are calculated from the last 24 hours. Min and Max include their exact timestamps so you can correlate an extreme with a switch event, weather change, irrigation cycle, or equipment problem. Consequently, a six-hour tile can show a 24-hour minimum or maximum that lies outside the six hours drawn in the tile.

Graph Background Colors And Thresholds

Colored backgrounds use that metric's gauge zones. The colors are metric-specific, not one universal alarm scale. For example:

- Temperature moves from blue for the coldest range, through light blue and green, to yellow and red for hotter ranges.
- Relative humidity uses brown and yellow for dry ranges, light blue for the middle range, and progressively darker blue for wetter ranges.
- CO2 uses green for its central configured range, yellow on either side, and red at the configured extremes.
- AQI progresses from green through yellow, orange, red, purple, and maroon as the index rises.
- VPD uses several moisture-demand bands. The compact tile uses the metric's gauge zones; the full-screen VPD graph uses its dedicated VPD background bands, so its palette is not identical to every compact tile.
- A single-color background, such as barometric pressure's light blue or a light metric's yellow, identifies the metric's scale and does not by itself indicate an alarm.

The boundaries come from Sensorius's gauge-zone configuration for each metric. Changing a zone boundary in that configuration changes where the corresponding background color begins and ends. The standard UI currently exposes gauge size and display style, but not gauge-zone boundary editing. An automation's **Threshold** and **Hysteresis** control when its rule runs; they do not change graph or gauge colors.

Metric Ordering

The system-wide **Metric Set** in **General Settings > Display** controls the initial row state. In **Pick 6** mode, the dashboard follows **Metric 1** through **Metric 6** exactly from left to right and initially collapses each sensor after those six cards. Use the triangle beside the connection indicator to reveal or hide the sensor's remaining known metrics. Sensors with six or fewer metrics do not show the triangle. You can therefore establish the operational summary order in **Sensor Settings**. Factory defaults are selected by sensor type and generally put the device's primary measurement first: for example, CO2 is first for a CO2 sensor and Air Quality is first for an AQI sensor. The remaining positions favor closely related temperature, humidity, VPD, dew-risk, pressure, plant, light, or soil measurements for that device.

In **All** mode, Sensorius uses the same ordering but initially expands every sensor. The row triangle can still collapse the display back to its six-card summary. Sensorius does not apply a universal rule that moves CO2, AQI, and every other specialized metric farther right. Nor does it guarantee that barometric pressure is always fifth or that dew point fills the fifth position when pressure is unavailable. Assign **Metric 1-6** when an exact summary convention is important.

Sensor Names, Raspberry Pi Buses, And Locations

A directly connected Raspberry Pi sensor ID has the form `<kind>-<bus>-<hostname>`, for example `avpd-i2c-1-sensorius-demo`. The kind identifies the sensor family, the bus segment identifies the Linux I2C interface used during discovery, and the final segment identifies the hub. `i2c-1` is the normal sensor bus on GPIO2/GPIO3. `i2c-0` is the secondary bus on GPIO0/GPIO1 used for supported plant-probe arrangements. A dual-bus APVPD device is represented by one sensor ID using its primary `i2c-1` descriptor even though its plant probe also uses `i2c-0`.

The technical sensor or switch ID remains the stable identity used by settings, database history, MQTT, and switch keys. **Location** is the friendly, editable place name used to group and filter dashboard cards and to make automation selectors understandable. A card header shows both: use the ID when diagnosing wiring, topics, or stored settings, and use the location when operating the system. Renaming a location does not rename the device or disconnect its history.

Switch Cards

Switch cards show local Raspberry Pi relay channels and remote Nodus switch channels through the same interface. Each channel has a label, current state, and recent state changes. Manual toggles send commands through the shared switch controller. For Nodus switches, commands are sent through MQTT and Sensorius waits for state to return from the device.

If an enabled Advanced automation owns a switch channel, Sensorius blocks manual toggles for that channel so the automation remains in control.

Switch state history comes from `sensorius.saiDataLogger.log_switch_event`, not from synthetic sensor readings. This is why switch events can be overlaid on historical graphs.

Each switch channel also has an auto-off timer. Open the timer control with its gear, enter `0` to disable it or 30-9999 seconds, and click **Ok**. The dashboard accepts values in 30-second steps. The setting is runtime state, not persistent switch configuration, so it must be set again after Sensorius restarts. When a manually controlled channel turns on, the countdown starts; at expiry Sensorius sends one Off command and records a timer-originated switch event. Turning the channel off clears the active countdown. Automation-originated state changes do not start the manual auto-off countdown, and an enabled Advanced automation must be disabled before a manual dashboard toggle is allowed.

The Events column shows up to five recent On/Off transitions, newest first, with local 12-hour AM/PM timestamps and a **manual** or **auto** origin when one is known. The list combines persisted switch events with live state updates and refreshes as commands and device reports arrive. Those same persisted events can be selected as vertical overlays in the full-screen graph.

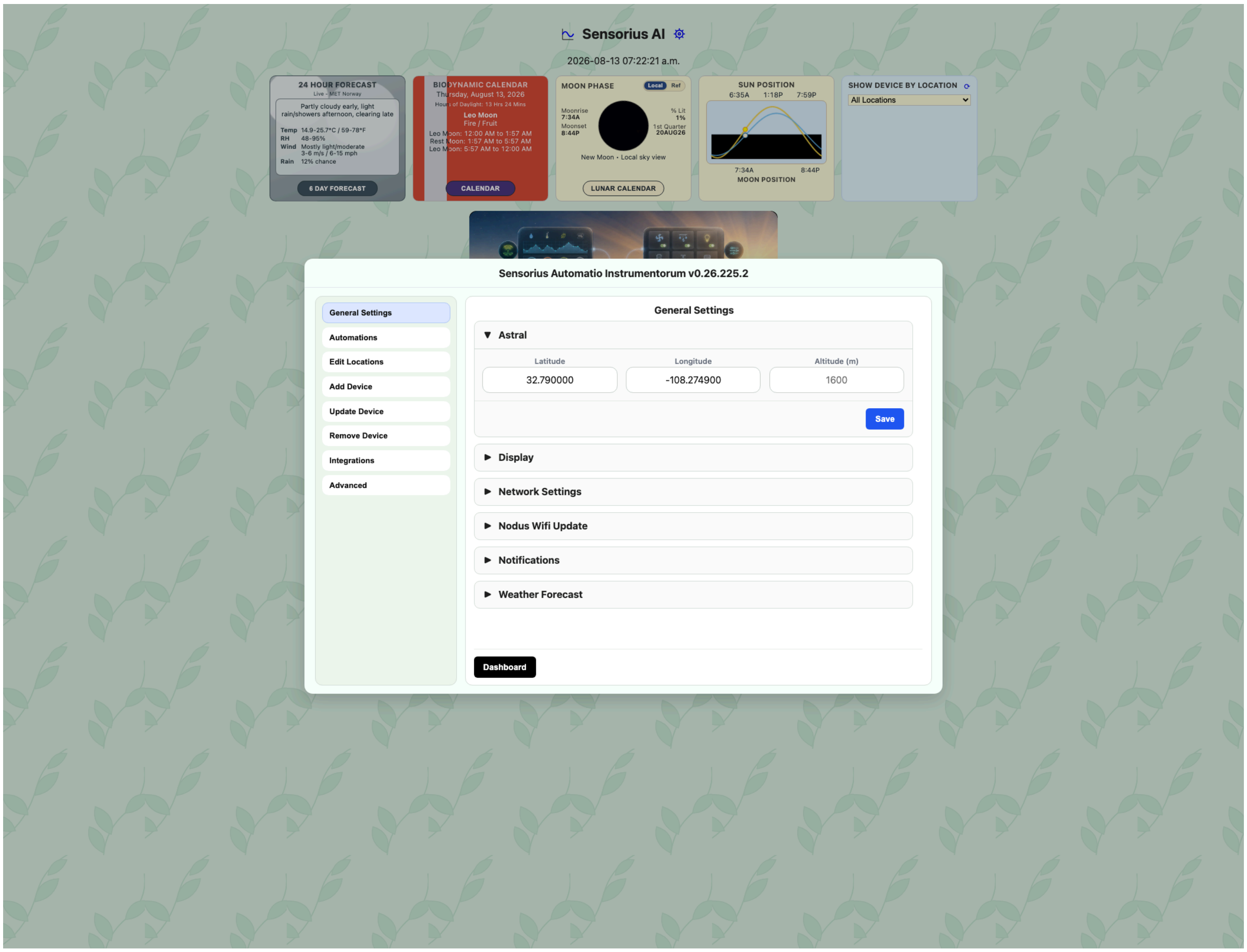
General Settings

General Settings contains hub-level settings, notifications, system-wide automations, device onboarding, integrations, locations, firmware updates, and maintenance tools.

General Settings Pane

The main pane contains six expandable sections. Open one to review or change its settings. Each section-level **Save** submits and writes only the controls inside that section; values in the other expandable sections are left unchanged.

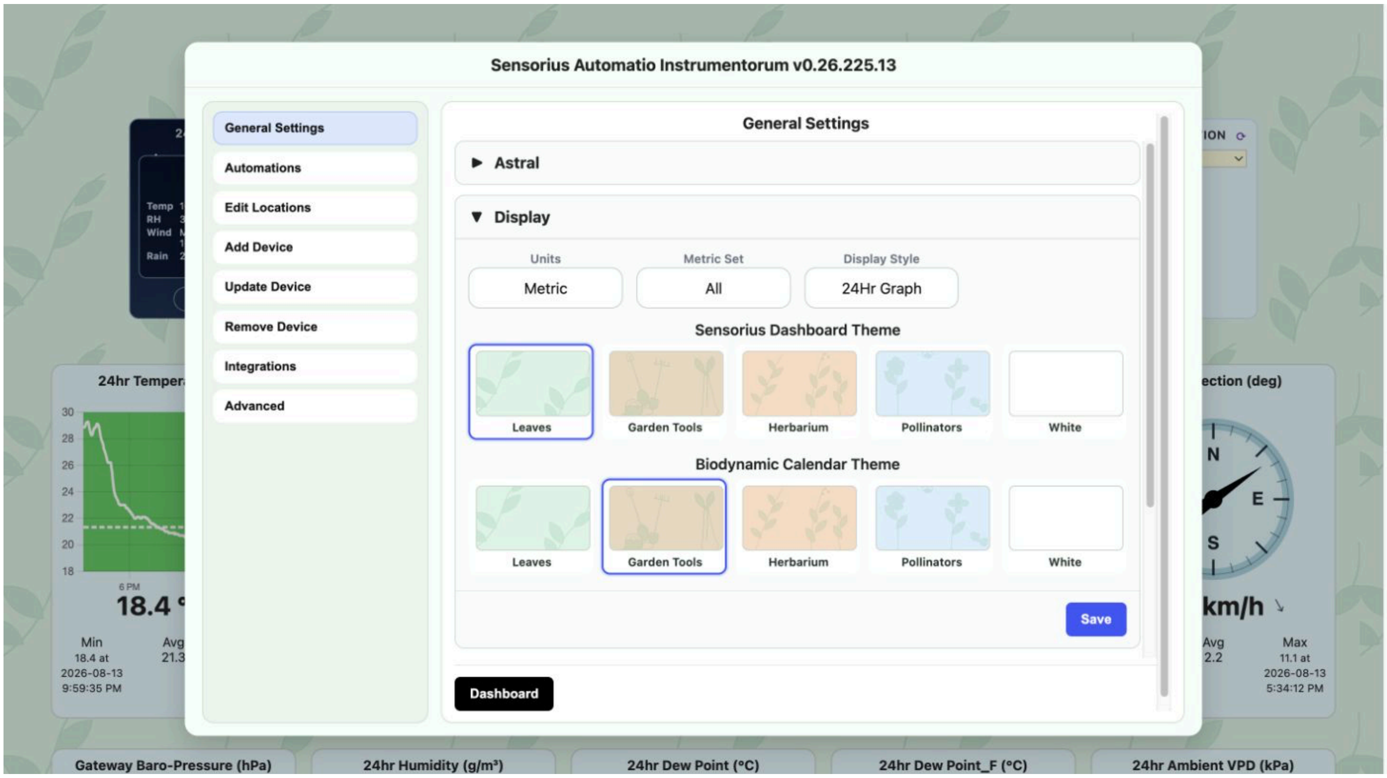
Astral



The **Astral** section sets the physical location used for sunrise, sunset, moon, biodynamic, weather, and location-aware automation calculations:

- **Latitude**: leave both Latitude and Longitude empty, then select **Save**, to detect the location automatically.
- **Longitude**: Latitude and Longitude must both be supplied when entering a location manually.
- **Altitude (m)**: altitude in meters. Valid values range from -500 to 10000.
- **Save**: writes the Astral location. Changing it causes dependent calendar and astronomy information to be recalculated.

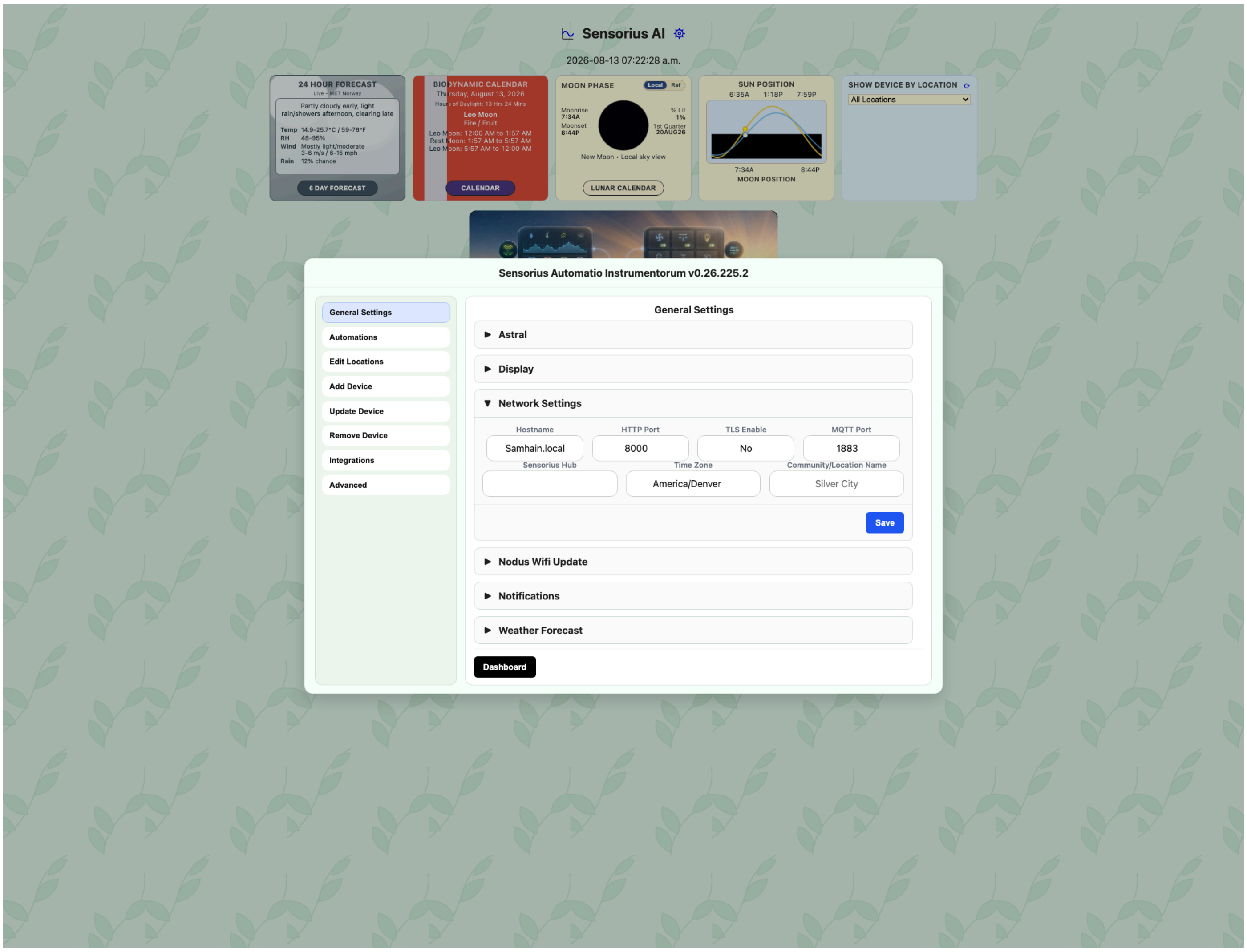
Display



The **Display** section supplies system-wide presentation defaults. The controls appear in the order **Units**, **Metric Set**, and **Display Style**. **Gauge Size** is shown only while **Gauge** is selected.

- **Units**: chooses **Imperial** or **Metric** presentation throughout the dashboard, historical graphs, and Caelus weather forecast. Sensorius converts temperatures, pressure, speed, rainfall, rain rate, and distance when the source unit has a supported equivalent. For example, Imperial displays use °F, inHg, mph, inches, and miles; Metric displays use °C, hPa, km/h, millimeters, and kilometers. Unitless values and measurements without an applicable conversion, such as relative humidity, VPD, CO2, and light, keep their native units. This setting changes labels, values, graph scales, and display zones only. It does not rewrite sensor readings, database history, MQTT payloads, sensor configuration, metric identities, or automation thresholds.
- **Metric Set**: applies to every sensor on the Sensorius dashboard. **Pick 6** initially shows each sensor's six saved metric slots; its row triangle reveals every additional known metric. **All** starts with those additional metrics expanded. Neither option changes sensor settings. Additional metrics use the system **Display Style**.
- **Display Style**: default metric display when a sensor has no saved per-metric style. Options are **Gauge**, **6Hr Graph**, and **24Hr Graph**.
- **Gauge Size**: dashboard gauge size. Options are **Small** and **Large**.
- **Sensorius Dashboard Theme**: five thumbnails select **Leaves**, **Garden Tools**, **Herbarium**, **Pollinators**, or **White** for the main dashboard.
- **Biodynamic Calendar Theme**: independently selects the full-screen calendar background from the same five choices. Both theme settings default to **Leaves**; **White** removes the repeating SVG background image.
- **Save**: writes the display defaults. Reload the dashboard to see changes that are not applied immediately.

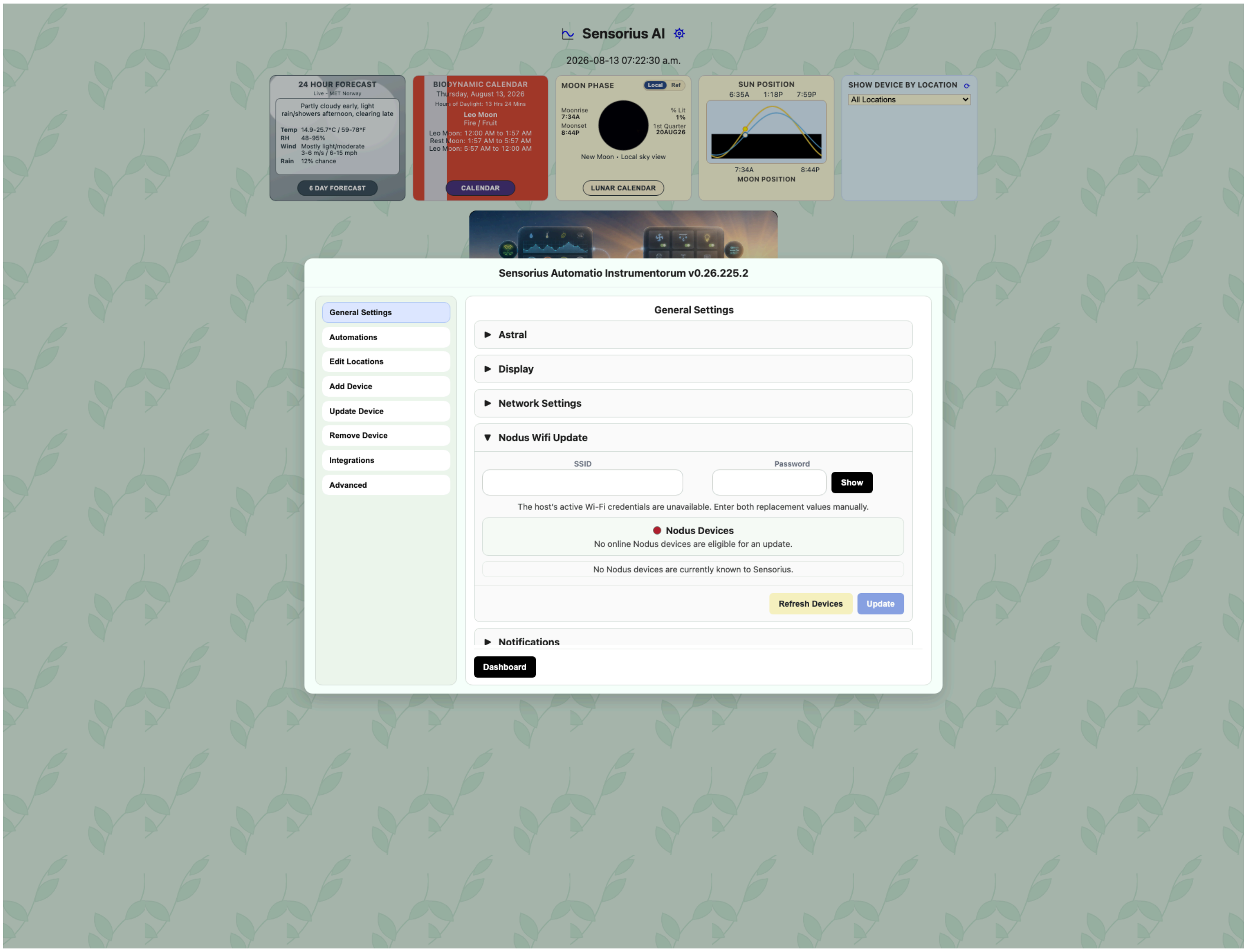
Network Settings



The **Network Settings** section contains the hub and primary MQTT connection settings:

- **Hostname**: read-only host name for this Sensorius hub. It comes from the active system settings and host runtime.
- **HTTP Port**: web UI port. Valid range is 1 to 65535. The default is usually 8000. Changing it may require a restart or opening the new URL.
- **TLS Enable**: MQTT TLS setting for the Sensorius sensor network. Options are **No** and **Yes**. Use Yes only when the broker is configured for TLS.
- **MQTT Port**: MQTT broker port. Valid range is 1 to 65535. Common values are 1883 without TLS and 8883 with TLS.
- **Sensorius Hub**: MQTT broker hostname or IP used by Sensorius and Nodus devices.
- **Time Zone**: IANA timezone name, such as `America/Denver`. It controls dashboard time labels, graph time labels, Astral timing, and calendar day boundaries.
- **Community/Location Name**: optional friendly community or locality shown as the large location heading in Caelus, such as `Silver City`. It does not change the Astral coordinates, timezone, or any sensor/switch location. Leaving it blank uses the default Caelus station label.
- **Dashboard**: returns to the dashboard.
- **Save**: writes the values in this section.

Nodus Wifi Update



Open the **Nodus Wifi Update** section in the main **General Settings** pane to send a replacement network name and password to connected Nodus devices before changing the router.

Before You Begin

- Keep the router on its existing Wi-Fi credentials until Sensorius finishes the update and reports the result for every expected Nodus.
- Power on every Nodus you intend to update and confirm that each one appears as **online**. An offline Nodus cannot receive the new credentials.
- Have the router administration password available. Also be prepared to reconnect the Sensorius host computer to the replacement Wi-Fi network after the router restarts.

Update Procedure

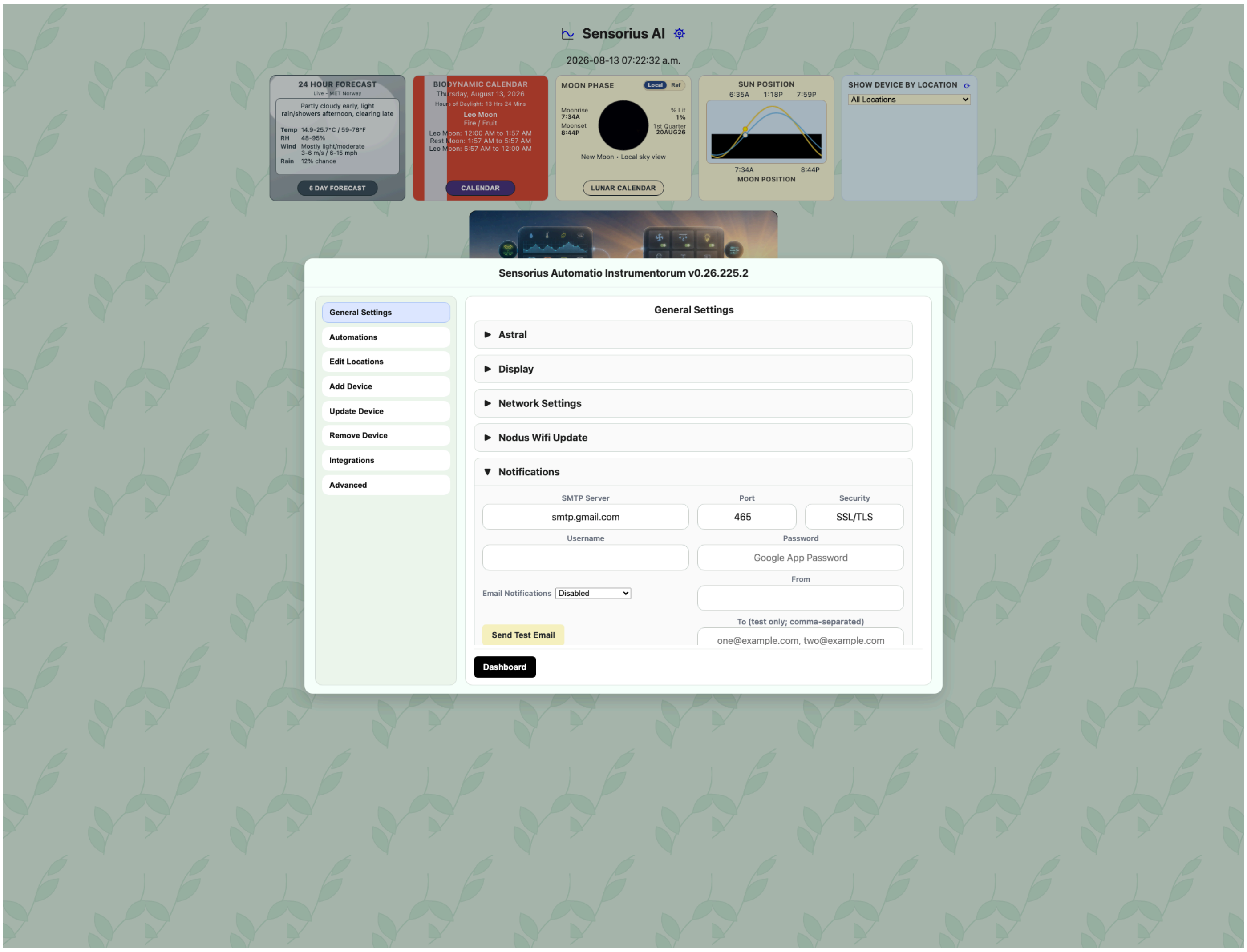
1. Open **General Settings**, then expand **Nodus Wifi Update**.
2. Wait for the device scan to finish. Check that every Nodus you expect to update is listed as **online**. Resolve missing or offline devices before continuing.

3. Review the **SSID** and **Password** fields. Sensorius attempts to populate them with the host computer's current Wi-Fi credentials when the operating system permits access. The password is masked; select **Show** to inspect it and **Hide** to mask it again. Enter the credentials manually if either field could not be read.
4. Replace the SSID, password, or both with the credentials the router will use after the change. These values are case-sensitive.
5. Select **Update**, review the confirmation list, and confirm the update. **Do not change or restart the router yet.**
6. Sensorius sends the SSID and password one setting at a time to every eligible Nodus, waiting for each setting to be acknowledged and saved. It completes this staging pass before sending restart commands. Only devices that confirm both settings are restarted.
7. Review every per-device result:
   - **Restarting** means the new credentials were saved and the restart command was accepted.
   - **Failed** means the credentials were not confirmed.
   - **Restart failed** means the credentials were saved, but Sensorius could not confirm the restart command.
   - **Skipped** or **unavailable** means the device was not eligible for the update, commonly because it was offline.
8. When all expected devices report **Restarting**, sign in to the router, change its SSID and password to the same values, and allow it to reboot.
9. Reconnect the Sensorius host computer to the replacement Wi-Fi network. A host using a wired Ethernet connection may remain reachable throughout the change.
10. Reopen the Sensorius dashboard and verify that every Nodus returns online. A Nodus may enter AP mode while the router is unavailable and can take one AP restart cycle, typically about 5 to 10 minutes, to rejoin.

If the results are mixed, do not assume the fleet update is complete. Successful devices have already received the new credentials and may be restarting, while failed or skipped devices still use the old credentials. Retry any device that remains reachable, or recover it through Nodus AP mode, before completing the network cutover. After the router change, a Nodus that does not return online must be connected to and corrected through AP mode.

The host credentials are loaded transiently into the form. Both fields are cleared immediately after submission or when the settings window closes. Sensorius does not save either value in hub settings, the database, browser storage, logs, or metadata shadows. The credential-read response and the non-retained MQTT command necessarily carry the values in transit; enable MQTT TLS and protect access to the Sensorius web UI.

Notifications



Enable **Email Notifications** and save the SMTP settings before creating a **Notify** action. The automation editor shows **Notify** only while email notifications are enabled. Disabling email notifications hides that actor and prevents automated email delivery, but does not remove saved automations.

The section configures the SMTP server, port, TLS mode, username, App Password, sender address, and enabled state. **To** accepts one or more comma-separated addresses used only by **Send Test Email**. Automation recipients are set on individual **Notify** actions under **General Settings > Automations**. Email connection values are saved in the protected project-root `.env` file.

Configure Gmail for Sensorius

Sensorius sends Gmail messages through Google's authenticated SMTP service. Do not enter the normal password for the Google Account. Google requires an App Password for this type of connection, and App Passwords are available only after 2-Step Verification is enabled.

1. Sign in to the Google Account that Sensorius will use to send notifications. Using a dedicated account for the hub can make access and alert history easier to manage.

2. Open the Google Account **Security** page and enable **2-Step Verification** if it is not already enabled. Complete Google's enrollment and verification prompts.

3. Open Google's App Passwords page. Sign in again if prompted.

4. Enter a descriptive app name such as `Sensorius Hub` and select **Create**.

5. Copy the generated 16-character App Password immediately. Google displays it only once. Sensorius accepts it with or without the spaces shown by Google.

6. In Sensorius, open **General Settings > Notifications** and enter:

   - **SMTP Server**: `smtp.gmail.com`
   - **Port**: `465`
   - **Security**: **SSL/TLS**
   - **Username**: the complete Gmail address, such as `hub@example.com`
   - **Password**: the generated Google App Password, not the account password
   - **From**: normally the same complete Gmail address
   - **Email Notifications**: **Enabled**

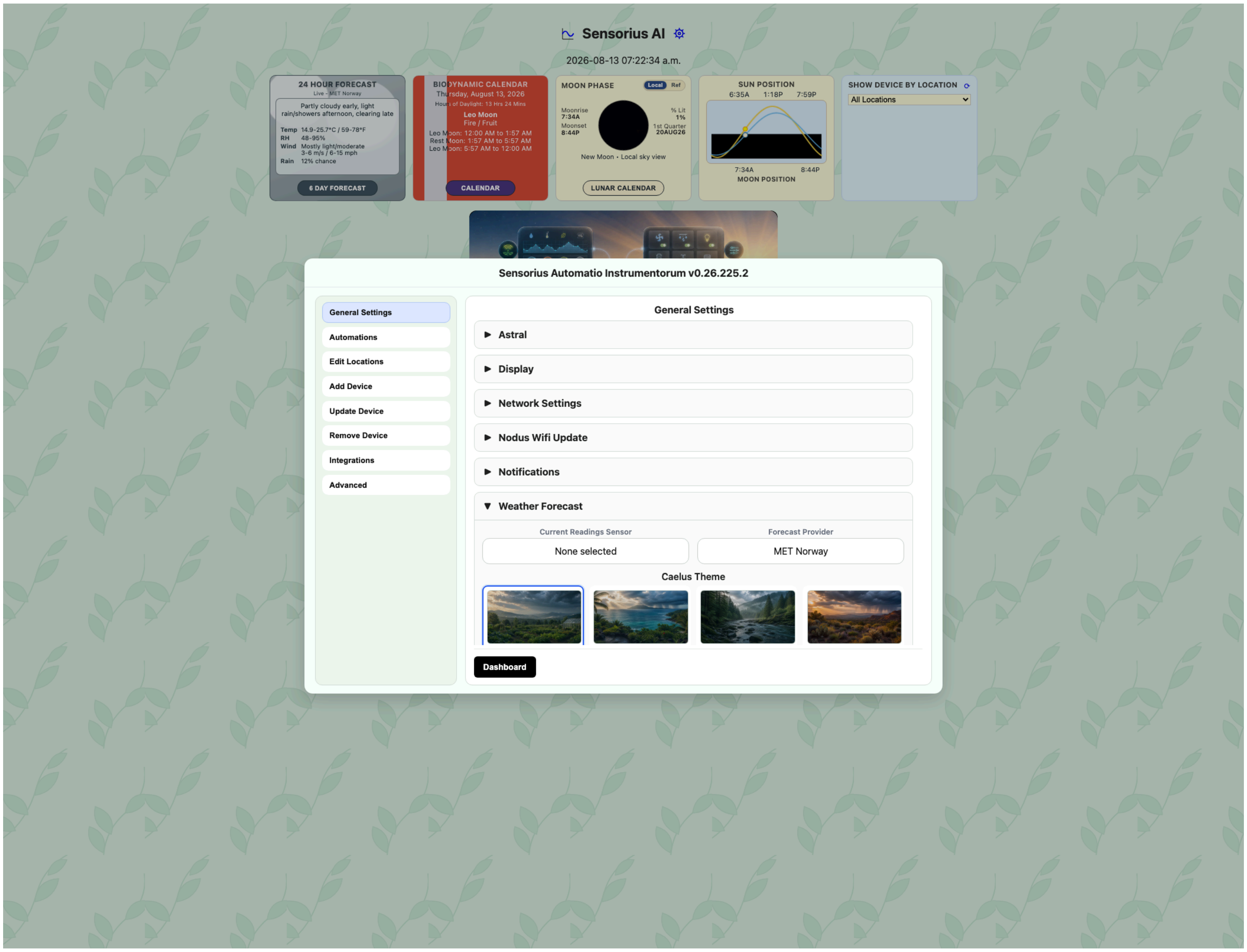
7. Enter one test recipient, or multiple comma-separated recipients, in **To**, then select **Send Test Email**. This tests the values currently displayed, so they do not have to be saved first. Check the recipient inbox and spam folder if Sensorius reports success but the message is not visible.

8. After the test succeeds, select **Save**. On later edits, leave **Password** blank to retain the stored App Password.

Port `587` with **STARTTLS** is also supported. Keep the port and security mode paired: use port `465` with **SSL/TLS**, or port `587` with **STARTTLS**. Google documents the SMTP host, authentication, and TLS requirements in its Gmail client settings, and explains App Password creation and restrictions in Google Account Help.

If **App Passwords** is unavailable, confirm that 2-Step Verification is fully enabled. Google may also hide App Passwords for work or school accounts managed by an organization, accounts enrolled in Advanced Protection, or accounts whose 2-Step Verification is configured only with security keys. For a managed Google Workspace account, ask the administrator whether authenticated SMTP and App Passwords are permitted. Google revokes existing App Passwords when the main Google Account password changes; create and save a new App Password if email delivery stops afterward. Revoke the Sensorius App Password from the Google Account when the hub is retired or no longer uses that account.

Weather Forecast

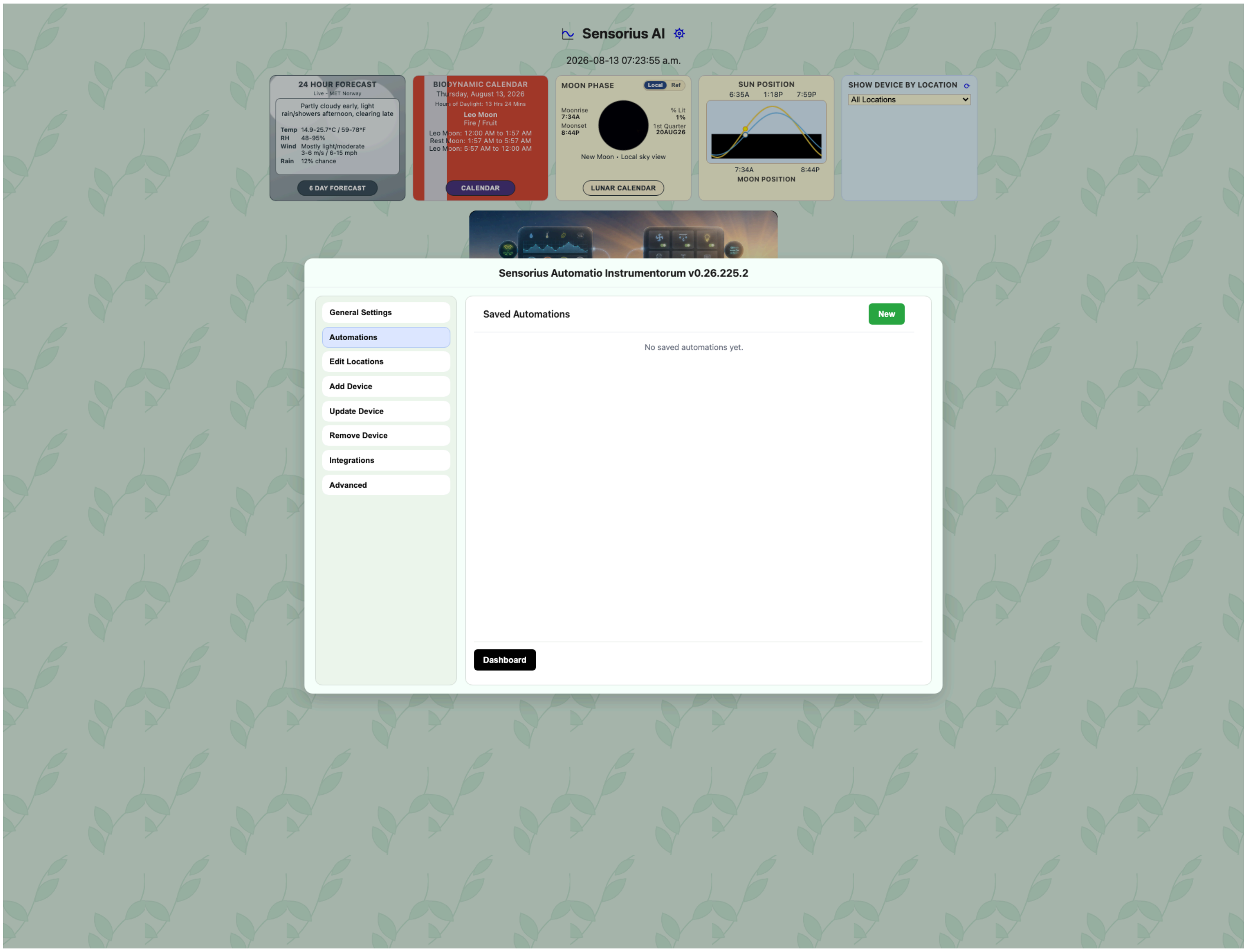


Fields and selectors:

- **Forecast Provider**: selects the forecast source used by the dashboard and Caelus. Options are **MET Norway**, **US · National Weather Service**, **Open-Meteo**, and **Disabled**.
- **Caelus Theme**: four image thumbnails select the full-screen scene: **Mountain Garden**, **Ocean Island**, **Forest River**, or **Desert Bloom**.
- **Current Readings Sensor**: selects any live sensor in the sensor directory. Directly connected Raspberry Pi sensors remain selectable while their first reading is being collected after startup. If the directory is briefly empty, the selector retries automatically and refreshes whenever Integrations is activated or reopened. Caelus displays that sensor's configured Display Metrics in their saved order. A WeeWX station can therefore supply outdoor temperature, humidity, rain, wind direction, and barometric pressure.
- **Save**: writes `[WeatherForecast]` provider, theme, and sensor selection.

Forecast placement, astronomy, sunrise, and sunset use the latitude, longitude, and timezone under **General Settings > Astral**. The Current Readings panel follows the selected sensor's configured **Display Metrics**.

Automations Pane



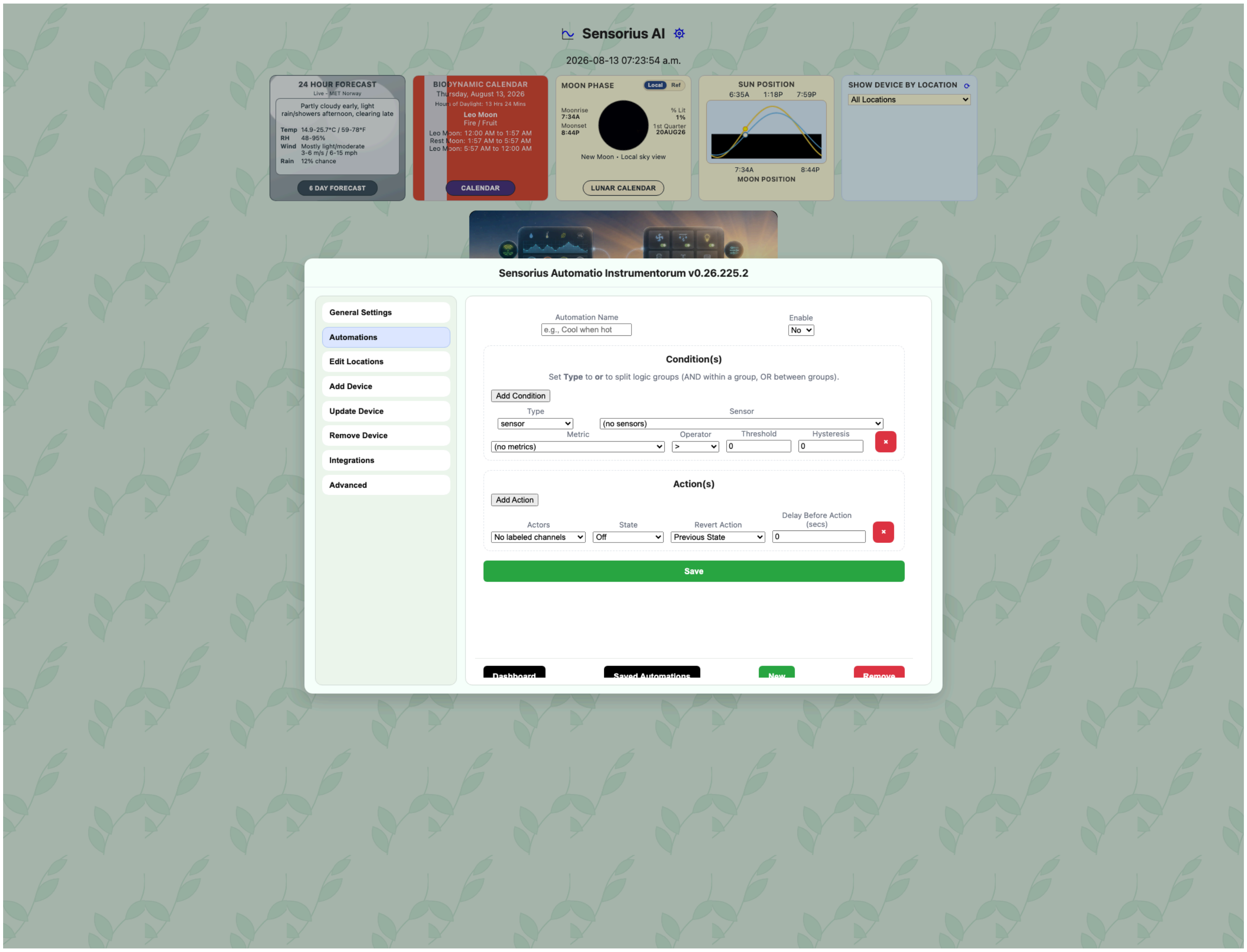
Open **General Settings > Automations**. Automations are configured globally from General Settings and are no longer edited from an individual switch's settings. The saved list shows every automation and whether it is enabled. This also allows a notification-only automation to run when no switch is installed.

Controls:

- **New**: opens a new automation definition.
- **Saved Automations**: returns from the editor to the saved list.
- **Remove**: deletes the selected automation.
- **Enable** in the editor: sets whether the rule is active. Options are **Yes** and **No**.

For compatibility, automation rules remain stored under the Sensorius runtime directory at `/Users/<user>/Sensorius/switch_settings/automations/automations.toml` on macOS or `/home/<user>/Sensorius/switch_settings/automations/automations.toml` on Linux. The General Settings editor loads all saved rules, sensor and metric choices, and the available actor directory.

Automation Definition



Top-level fields:

- **Automation Name**: friendly name shown in the saved automation list.
- **Enable**: **Yes** activates the automation, **No** saves it but does not run it.
- **Add Condition**: adds another condition row.
- **Add Action**: adds another action row.
- **Save**: saves the automation.

Condition **Type** options:

- **sensor**: compares a sensor metric to a threshold.
- **time of day**: active only inside a daily time window.
- **astral**: uses sunrise or sunset timing from Astral location.
- **timer**: active for a repeated duration, such as 5 minutes every hour.
- **or**: separates groups of conditions. Conditions within a group are AND; groups separated by OR are OR.

Sensor Conditions

- **Sensor**: sensor to watch. Options are dashboard-visible sensors.

- **Metric**: metric from the selected sensor.
- **Operator**: comparison operator. Options are `>`, `<`, `==`, and `!=`.
- **Threshold**: numeric value used by the comparison.
- **Hysteresis**: buffer around the threshold to reduce rapid on/off cycling.

Time-of-Day Conditions

- **Start Time**: beginning of the active window.
- **Stop Time**: end of the active window.
- **Days**: weekdays when the condition can be true. Options are Mon through Sun.

Astral Conditions

- **Event**: sunrise/sunset mode. Options are **sunrise to sunset**, **sunset to sunrise**, **sunrise**, and **sunset**.
- **Offset (minutes)**: shifts the event by -120 to 120 minutes. Negative starts before the event; positive starts after.
- **Days**: weekdays when the Astral rule can run.

Timer Conditions

- **Every**: repeat interval. Options are 5 minutes, 15 minutes, 30 minutes, 1 hour, 3 hours, 6 hours, 12 hours, and 24 hours.
- **Duration**: active duration in minutes, from 1 to 60. Duration must be less than the Every interval.

Actions

- **Actors**: action target. Switch entries are shown as `<switch_id>:<switch_label>` and use the stable channel ID behind the scenes. When email notifications are enabled, **Notify** is also available.
- **State**: shown for a switch actor; selects **On** or **Off**.
- **Revert Action**: shown for a switch actor. **Previous State** returns the switch to its previous state when the rule is no longer true. **Do Nothing** leaves the switch where the action put it.
- **Delay Before Action (secs)**: shown for a switch actor; waits 0 to 60 seconds after the rule becomes true before applying the action.
- **To**: shown for the **Notify** actor. Enter the recipient for this automation. Sensorius sends a **TRIGGERED** email when the automation becomes true and a **CLEARED** email when it becomes false. Each message lists the AND/OR condition groups and their results, current sensor values where applicable, and all configured actions. Delivery uses the global rolling hourly and 24-hour email limits and retries a failed SMTP attempt up to three times. If every attempt fails, connected dashboards display an error toast that remains visible until clicked. Sensorius records the state only after successful email delivery, preventing duplicate messages after a restart while still detecting a changed state after downtime.

Switch actions set absolute states, not toggle or invert commands. All actions in one automation share the same condition groups. **Previous State** is captured when an action actually changes a switch; if the switch is already at the requested state, there is no new previous state for that action to restore.

To alternate two switch channels in one timer automation:

1. Disable the automation.
2. Put the channels in their normal baseline state manually. For example, set Green **On** and Yellow **Off**.

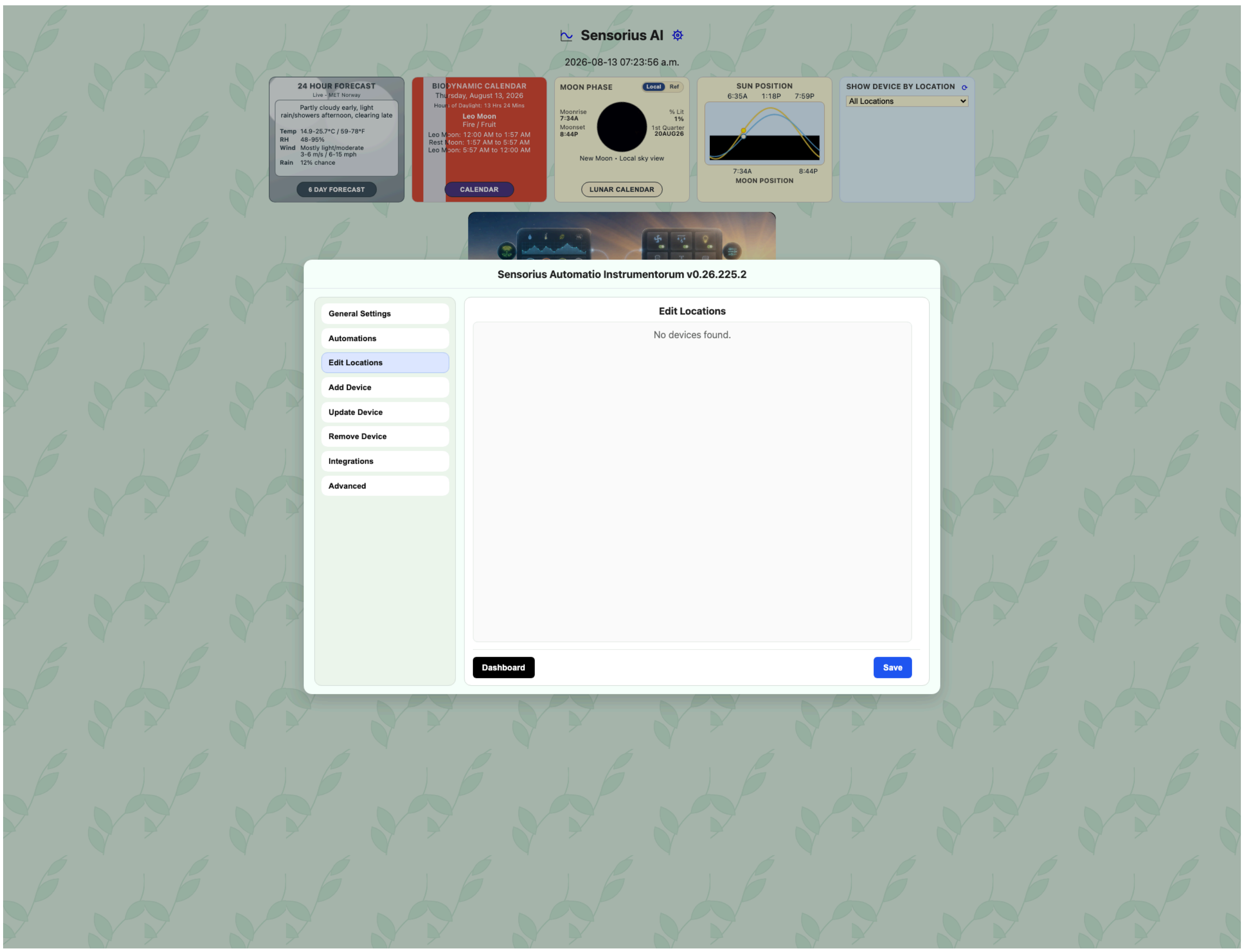
3. In the automation action rows, choose the state wanted during the timer window. For example, set Green **Off** and Yellow **On**.
4. Set **Revert Action** to **Previous State** for both rows.
5. Save and enable the automation.

With that setup, the timer window applies the action states, and the end of the window restores the manually set baseline state.

For critical loads, test automations with harmless equipment first. A short delay and hysteresis can prevent relay chatter when readings hover near a threshold.

If the Astral fields are wrong, biodynamic timing, sunrise/sunset automations, and weather forecast placement may also be wrong.

Edit Locations Pane



The Edit Locations pane lists sensors and switches together.

Fields:

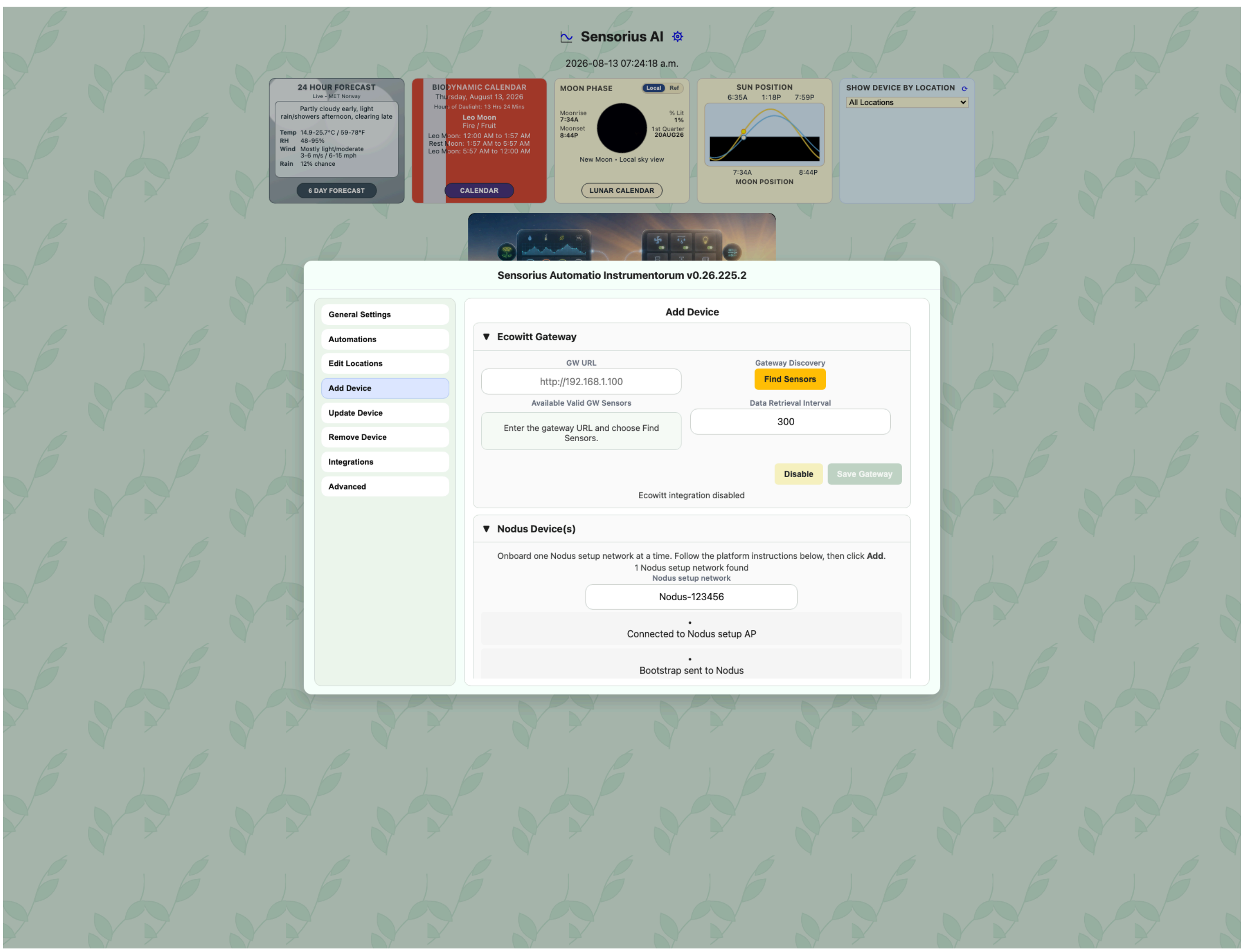
- **Device row label**: shows whether the row is a sensor or switch and shows its ID. Data comes from sensor and switch settings managers.
- **Location input**: updates that device's location. Sensor rows write to `sensor_settings/<sensor_id>/sensor.toml`; switch rows write to `switch_settings/<switch_id>/switch.toml`.
- **Save**: writes all non-empty location changes.

Locations should describe places people recognize: Greenhouse 1, West Bed, Seedling Bench, Main Pump, Hoop House, or Barn Weather Station. After a location change is saved, Sensorius reloads the dashboard so existing sensor row headings, switch-card grouping, and the location filter use the new value.

Add Device Pane

Use Add Device to onboard a factory-bootstrapped Nodus device or add a local Ecowitt gateway. The page groups device types into expandable sections. **Nodus Device(s)** contains the Nodus onboarding workflow. **Ecowitt Gateway** discovers and enables a read-only LAN weather-station connection.

Ecowitt Gateway

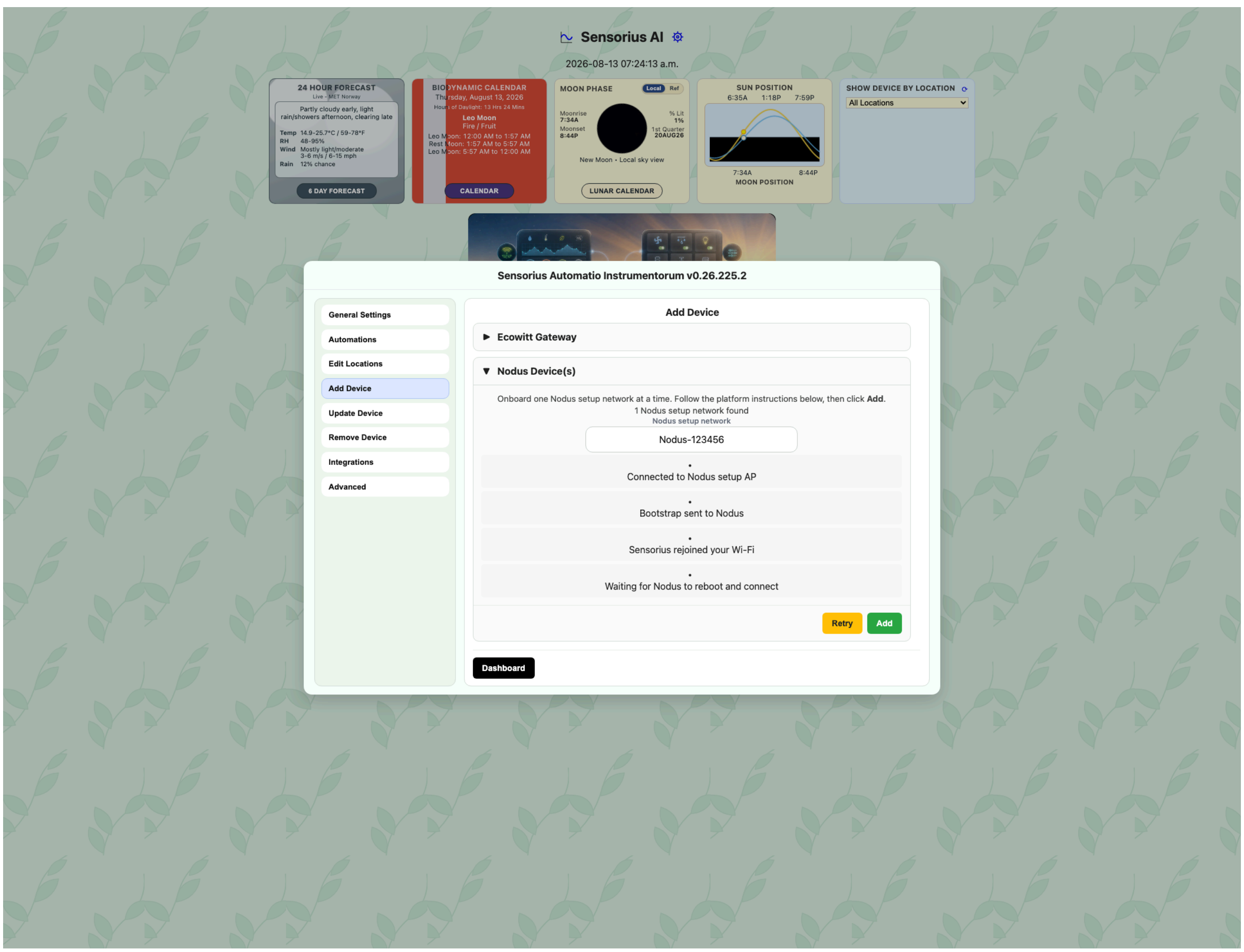


Ecowitt Gateway fields and controls are arranged in two columns:

- **GW URL**: the gateway base address, such as `http://192.168.1.100` or a local hostname. Do not include a path, credentials, query, or fragment.
- **Find Sensors**: queries the gateway's read-only version, network, sensor inventory, live-data, and rain-priority endpoints. Both inventory pages are checked.
- **Available Valid GW Sensors**: lists registered gateway sensors and whether their data family is present in the current live response. All valid listed sensors and supported additional channels are ingested.
- **Data Retrieval Interval**: polling period from 60 through 3600 seconds.
- **Save Gateway**: revalidates the gateway, derives a stable Sensorius station ID from its MAC address, creates station settings, and enables polling.
- **Disable**: stops polling without deleting station settings or historical readings.

Configure the GW1100 on the same trusted LAN first. A DHCP reservation is recommended. Sensorius accepts metric or imperial gateway display units and normalizes stored metric semantics. The local API's unit tags are authoritative; gateway-local unit settings can differ from Ecowitt app display preferences. Sensorius normalizes wind speed into its canonical mph metric, while wind direction drives the compass and 6/24-hour wind roses. As with WeeWX, that combined card's current reading and statistics show wind speed. Sensorius reads the gateway only; it does not change Wi-Fi, sensor registration, calibration, rain settings, MQTT, firmware, weather-service configuration, or gateway units.

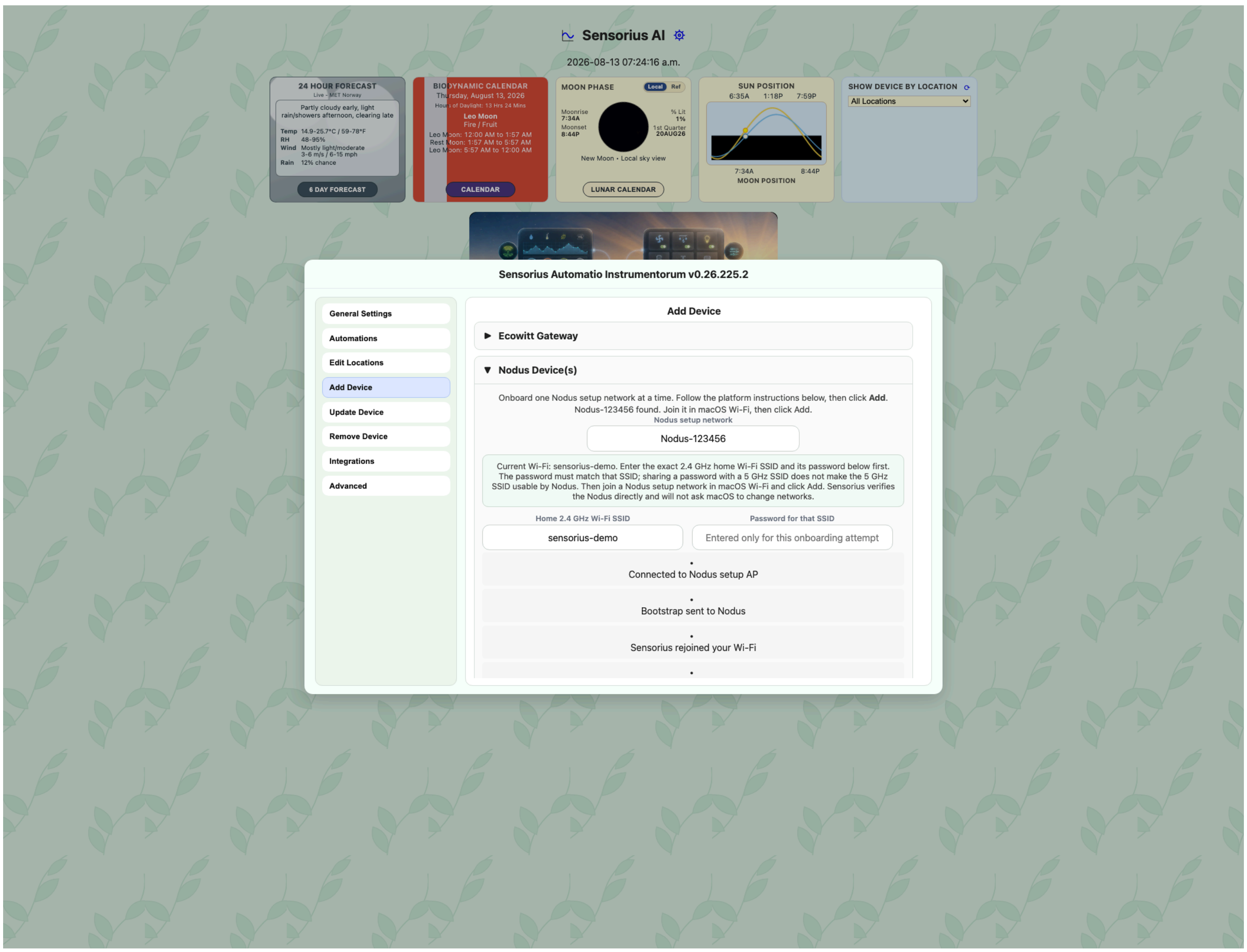
Nodus on Linux and Raspberry Pi



On Linux and Raspberry Pi, Sensorius scans for the Nodus setup access point, joins it, sends the bootstrap configuration, and restores the previous network. A failed scan leaves Add disabled; confirm that the Nodus AP is broadcasting, then use Retry. For Raspberry Pi deployments that onboard over Wi-Fi, use a 2.4 GHz network path. If the router combines 2.4 GHz and 5 GHz under one SSID, ethernet on the Raspberry Pi is usually the most reliable setup. If Add Device reports `network_control_not_authorized`, the Linux/Raspberry Pi service install did not grant NetworkManager control; use the Operations guide to repair the service authorization.

Before switching Wi-Fi, the page creates a resumable onboarding session. A browser connected through the Pi's Wi-Fi may be temporarily unreachable while the Pi joins the Nodus AP; progress polling resumes after the Pi restores its previous network. Run Sensorius through `sensorius.service`, not a foreground SSH process, during this transition.

Nodus on macOS



macOS requires administrator authorization when an application changes Wi-Fi networks programmatically and may request the administrator username and password multiple times during one onboarding attempt. Even after the user authorizes those changes, macOS can report stale or incomplete Wi-Fi state. That makes automatic switching less predictable and less user-friendly than a single manual network selection, so Sensorius deliberately leaves the Wi-Fi change to the user.

Enter the exact 2.4 GHz home Wi-Fi SSID and the password for that SSID in Add Device before changing networks. The SSID matters even when a router uses the same password for separate 2.4 GHz and 5 GHz networks; select the 2.4 GHz network because Nodus cannot join a 5 GHz-only SSID. Join the `Nodus-<serial-number>` setup network in macOS Wi-Fi, then return to Sensorius and click Add. A previously used Nodus may appear under Known Networks; a new one may appear under Other.

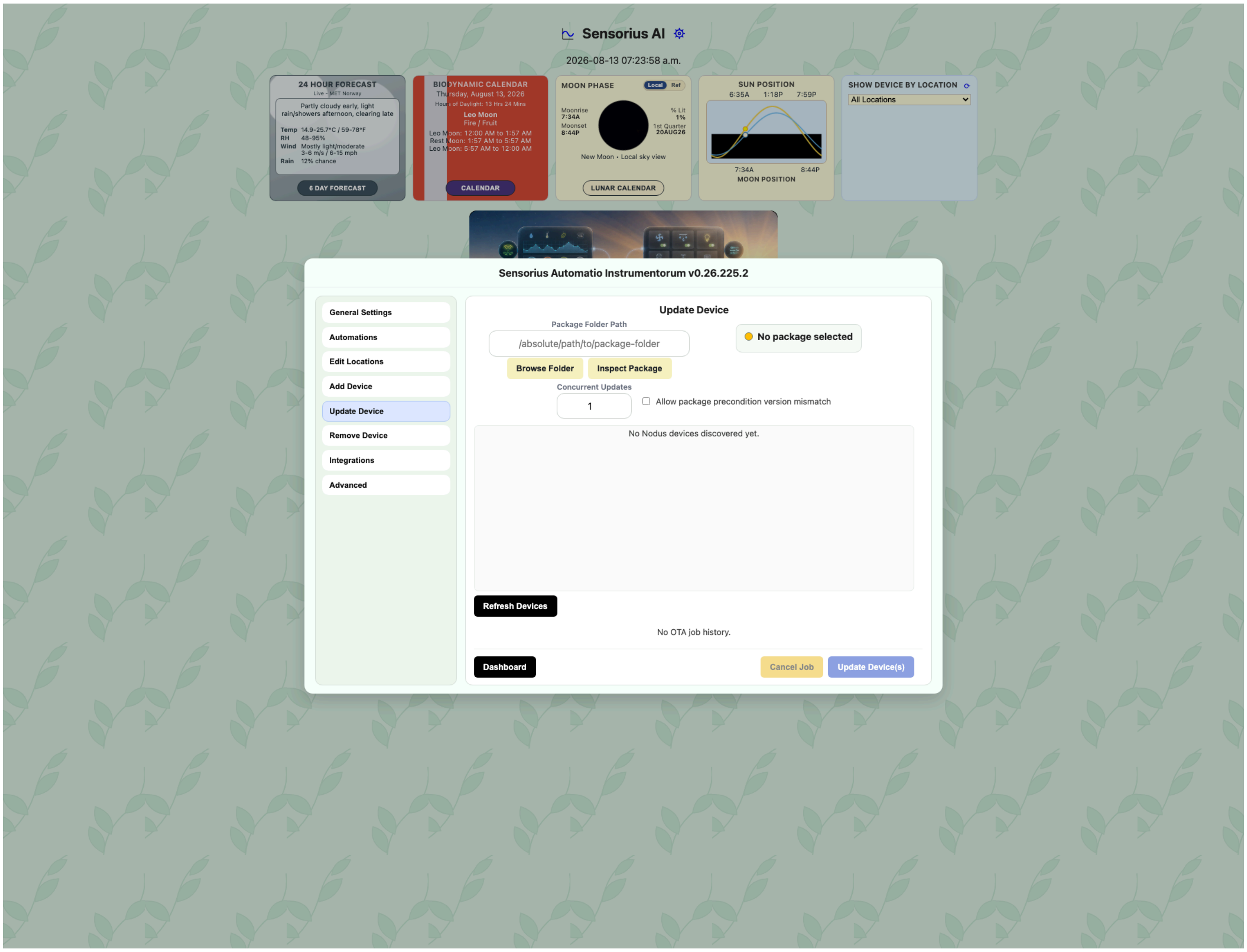
Sensorius verifies the manual join using the `192.168.4.x` Wi-Fi address and the Nodus `/itaot-meta` endpoint. It does not use `networksetup` to join, inspect, or restore Wi-Fi because current macOS versions can report error `-3900` or stale association state after a successful graphical join. After bootstrap, rejoin the home network in macOS Wi-Fi if it does not reconnect automatically. Ethernet is not required: when Wi-Fi is the Mac's only network connection, keep Sensorius open at `http://127.0.0.1:8000`; the Nodus AP disappears after bootstrap and macOS can then reconnect to the home network. Sensorius waits for the MQTT handshake and the Nodus retries its broker connection until the Mac is reachable. If the AP disappears before its HTTP response reaches Sensorius, Add Device reports that the response was lost and continues waiting for the correlated MQTT hello instead of immediately declaring failure.

Shared Nodus fields and status rows:

- **Scanning for Nodus_Setup**: Sensorius scans for the Nodus setup access point.
- **Connected to Nodus setup AP**: confirms Sensorius or the host joined the Nodus setup network.
- **Bootstrap sent to Nodus**: confirms Wi-Fi and broker bootstrap information was sent.
- **Sensorius rejoined your Wi-Fi**: confirms the hub returned to the normal network.
- **Waiting for Nodus to reboot and connect**: waits for the Nodus device to reboot, join Wi-Fi, connect to MQTT, and send its hello/config result.
- **Retry**: retries the current onboarding session.
- **Add**: starts onboarding after the platform-specific setup steps are complete.

The legacy `Nodus_Setup` and `Nodus-Setup` names remain supported. When onboarding reaches Device Online, Sensorius automatically reloads the dashboard so the newly discovered sensor and switch cards appear without a manual browser refresh.

Update Device Pane



Use Update Device for Nodus over-the-air (OTA) firmware packages.

OTA has been verified for Nodus packages targeting `pico2w` and `xesp32s3`. Packages must declare the correct target platform; Sensorius blocks updates when the package target does not match known device metadata. For Pico 2 W, avoid large single-file compiled updates: command-line OTA testing showed a Nodus-side memory allocation failure while transferring an `app.mpy` larger than about 50 KB.

cPyNodus III package folders use signed `nodus-ota/v2`. Sensorius displays them as authenticated only after verifying `manifest.sig` with an operator-installed trusted public key. Existing cPyNodus II `nodus-ota/v1` package folders remain supported without a cPyNodus II firmware change.

Future Sensorius releases may accept zip uploads, but the current OTA workflow expects the package folder produced by `cPyNodus_II` or `cPyNodus_III`.

Fields and controls:

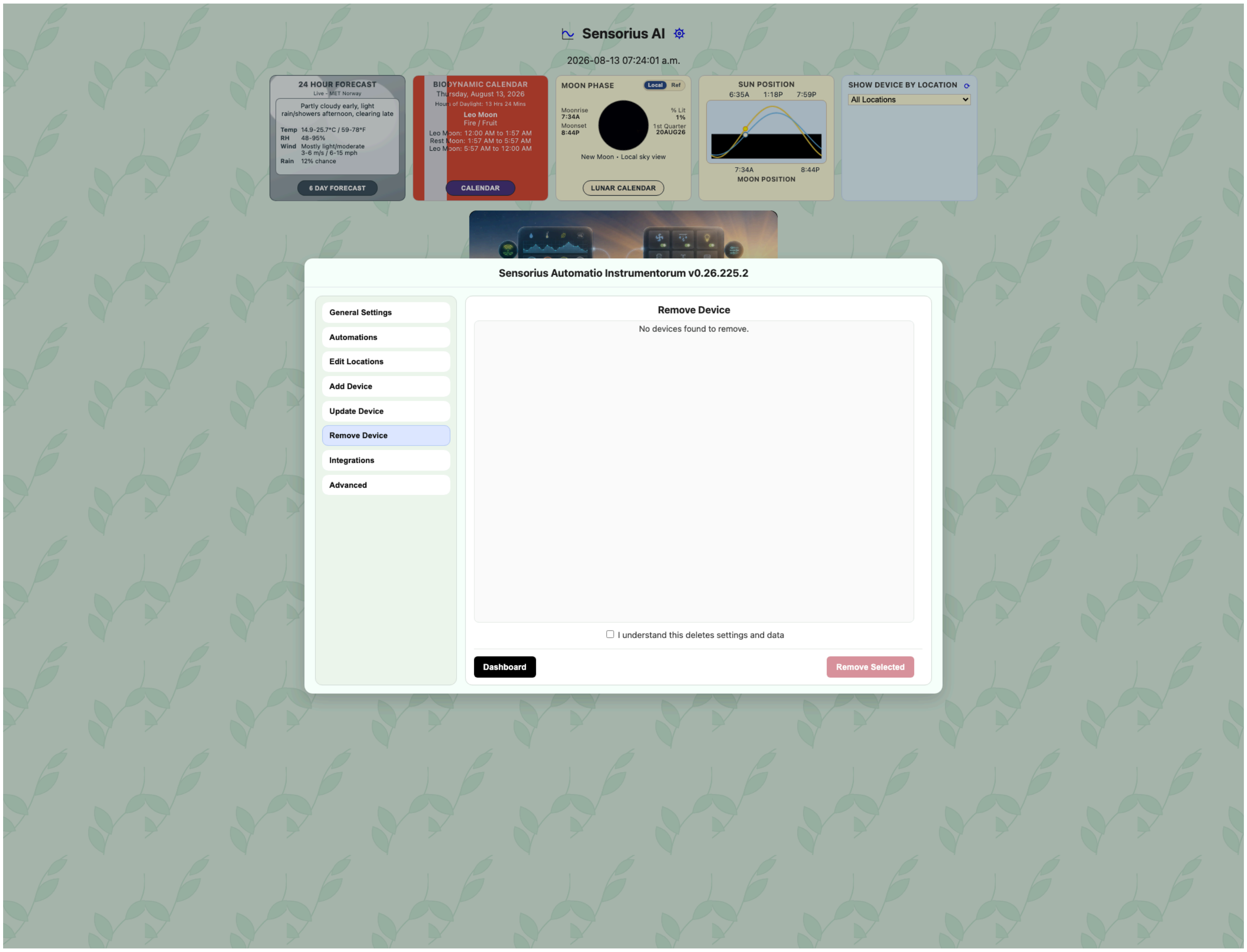
- **Package Folder**: local OTA package folder produced by `cPyNodus_II` or `cPyNodus_III`, selected from the Sensorius host.

- **Package summary**: package inspection status, protocol, authentication state, and signing key ID after selection or inspection.
- **Package Path**: absolute path to a package on the Sensorius host.
- **Inspect Package**: reads package metadata and validates that Sensorius can use it.
- **Concurrent Updates**: number of devices to update at once. Options are numeric values from 1 to 4.
- **Device list**: Nodus devices available for update. Data comes from known Nodus metadata and OTA endpoints.
- **Refresh Devices**: reloads the device list.
- **Job panel**: active update status and progress.
- **Job history**: recent update result information.
- **Cancel Job**: cancels an active update job when available.
- **Update Device(s)**: starts an OTA update for selected devices after a valid package and target selection are available.

Only update devices when power and network are stable.

After prepare, `Nodus OTA mode booting...` may remain visible for up to 150 seconds while the device reboots and reconnects. Routine connection failures during this interval are expected and are not shown. Sensorius makes up to three attempts per file and stops one device update after 30 minutes. If OTA mode does not start or those limits are exhausted, Sensorius requests OTA abort and reports a concise recovery message.

Remove Device Pane



Use Remove Device when a sensor or switch should no longer appear in Sensorius.

Fields and controls:

- **Device checkbox list**: removable devices known from settings, discovery, and runtime state.
- **Device detail**: may show URL or last-seen age when known.
- **I understand this deletes settings and data**: required confirmation checkbox.
- **Remove Selected**: treats a selected sensor, switch, or both entries from the same Nodus as one physical-device removal. It deletes the complete sensor-and-switch identity family, settings, and related local data; clears runtime caches and retained MQTT/Home Assistant topics; suppresses retained replay for every related identity; and verifies that the device no longer appears.

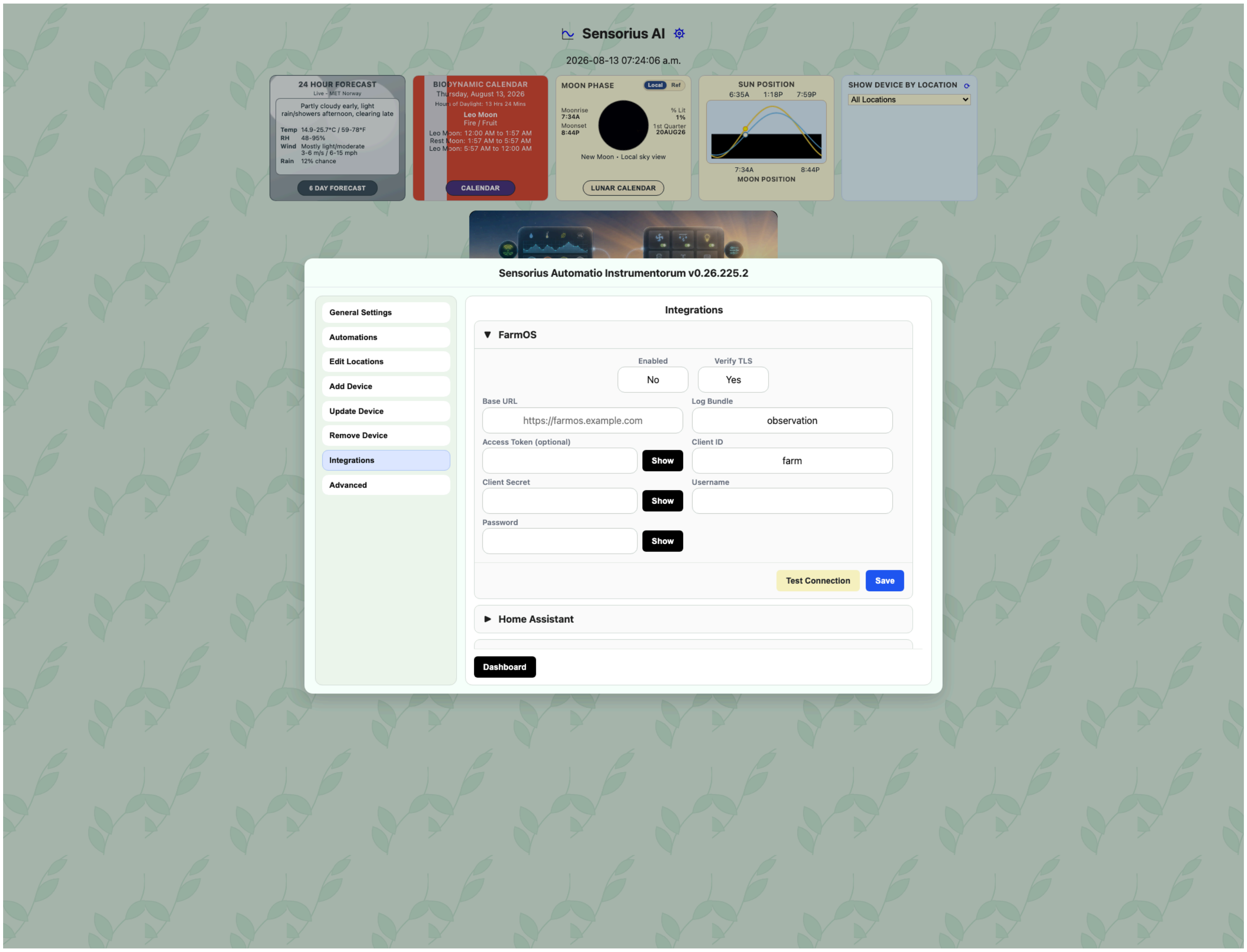
Removed Nodus identities remain ignored if retained or newly arriving MQTT messages are received later. Run Add Device onboarding again to intentionally allow and rediscover a removed Nodus device.

Before removing a device, update any automations, Home Assistant dashboards, farmOS expectations, or written operating procedures that depend on it.

Integrations Pane

General Settings presents Home Assistant, WeeWX, and FarmOS under the **Integrations** menu item. Each integration is an independently expandable block. Scroll the right pane vertically when the expanded blocks exceed the available height.

FarmOS



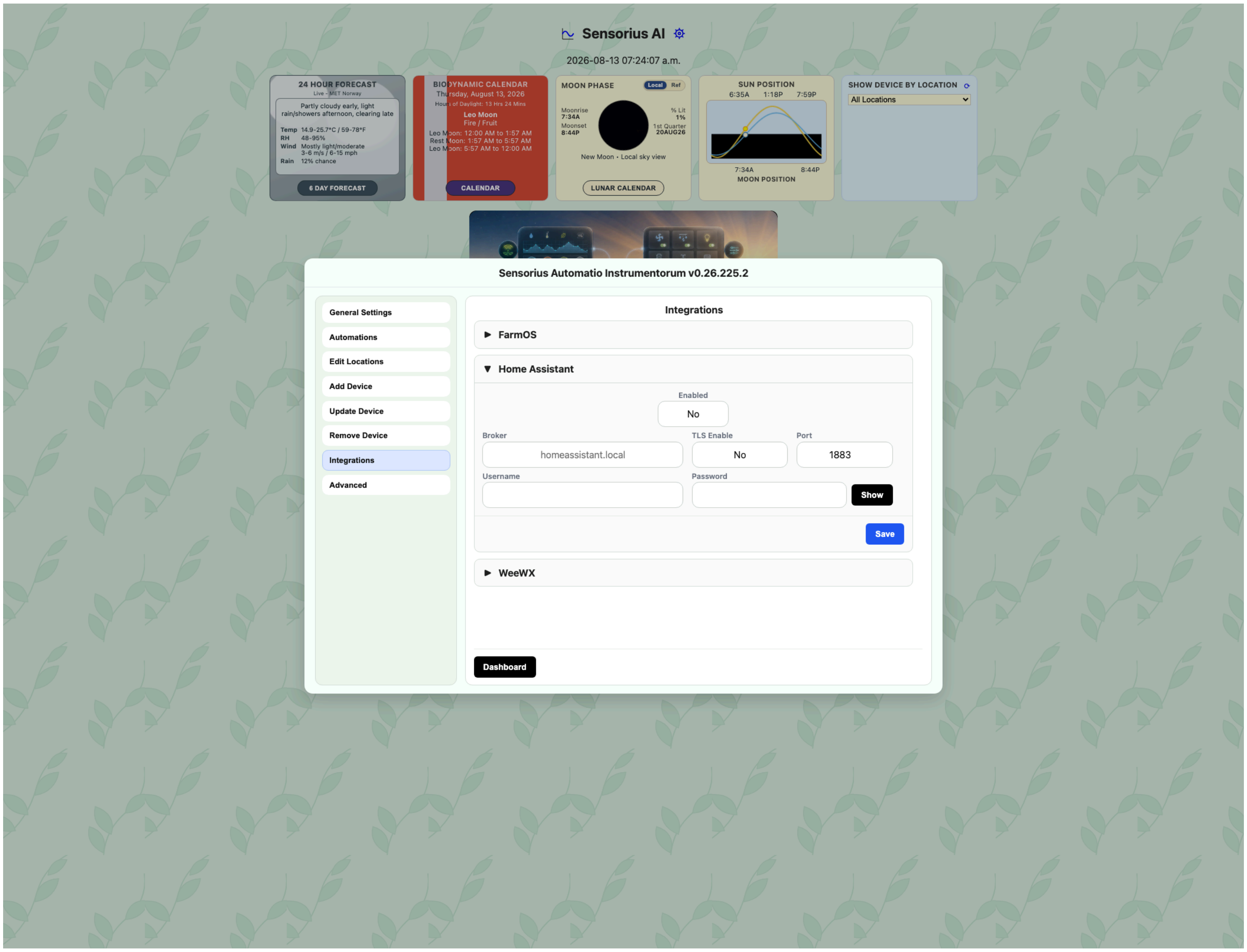
Fields and selectors:

- **Enabled**: turns farmOS export on or off. Options are **No** and **Yes**.
- **Verify TLS**: verifies the farmOS HTTPS certificate. Options are **Yes** and **No**. Leave Yes unless you are testing a private certificate.
- **Base URL**: farmOS site URL, such as `https://farmos.example.com`.
- **Log Bundle**: farmOS log bundle name. Default is `observation`.
- **Access Token (optional)**: static token, if you use token-based auth.
- **Client ID**: OAuth client ID. Default is `farm`.
- **Client Secret**: OAuth client secret.
- **Username**: farmOS username for password-based auth.
- **Password**: farmOS password for password-based auth.

- **Show** buttons: temporarily reveal hidden secret fields in the browser.
- **Test Connection**: calls the farmOS test endpoint and reports success or the last error.
- **Save**: writes the `[FarmOS]` settings.

farmOS export listens for new readings written by Sensorius. Check the FarmOS status if exports stop; it reports enabled state, queue depth, token state, and last error.

Home Assistant

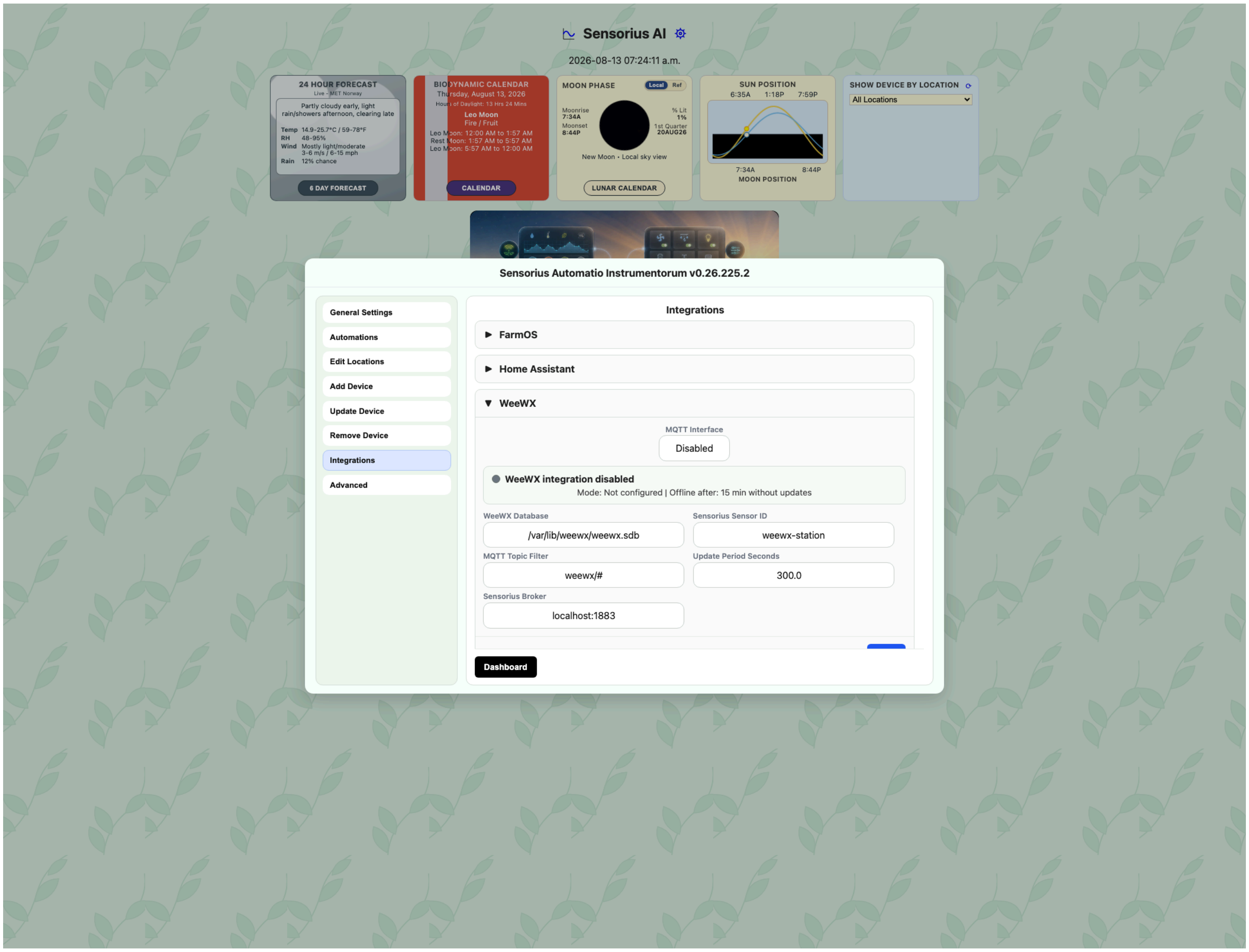


Fields and selectors:

- **Enabled**: turns Home Assistant integration on or off. Options are **No** and **Yes**.
- **Broker**: Home Assistant MQTT broker hostname or IP.
- **TLS Enable**: whether to use TLS for the Home Assistant broker. Options are **No** and **Yes**.
- **Port**: broker port, from 1 to 65535.
- **Username**: MQTT username if required.
- **Password**: MQTT password if required. Stored through Sensorius secret obfuscation.
- **Show**: temporarily reveals the password in the browser.
- **Save**: writes the `[HomeAssistant]` settings.

Expected flow: configure the broker, enable the integration, let Sensorius publish retained discovery topics, then let Home Assistant observe sensors and switches through MQTT.

WeeWX



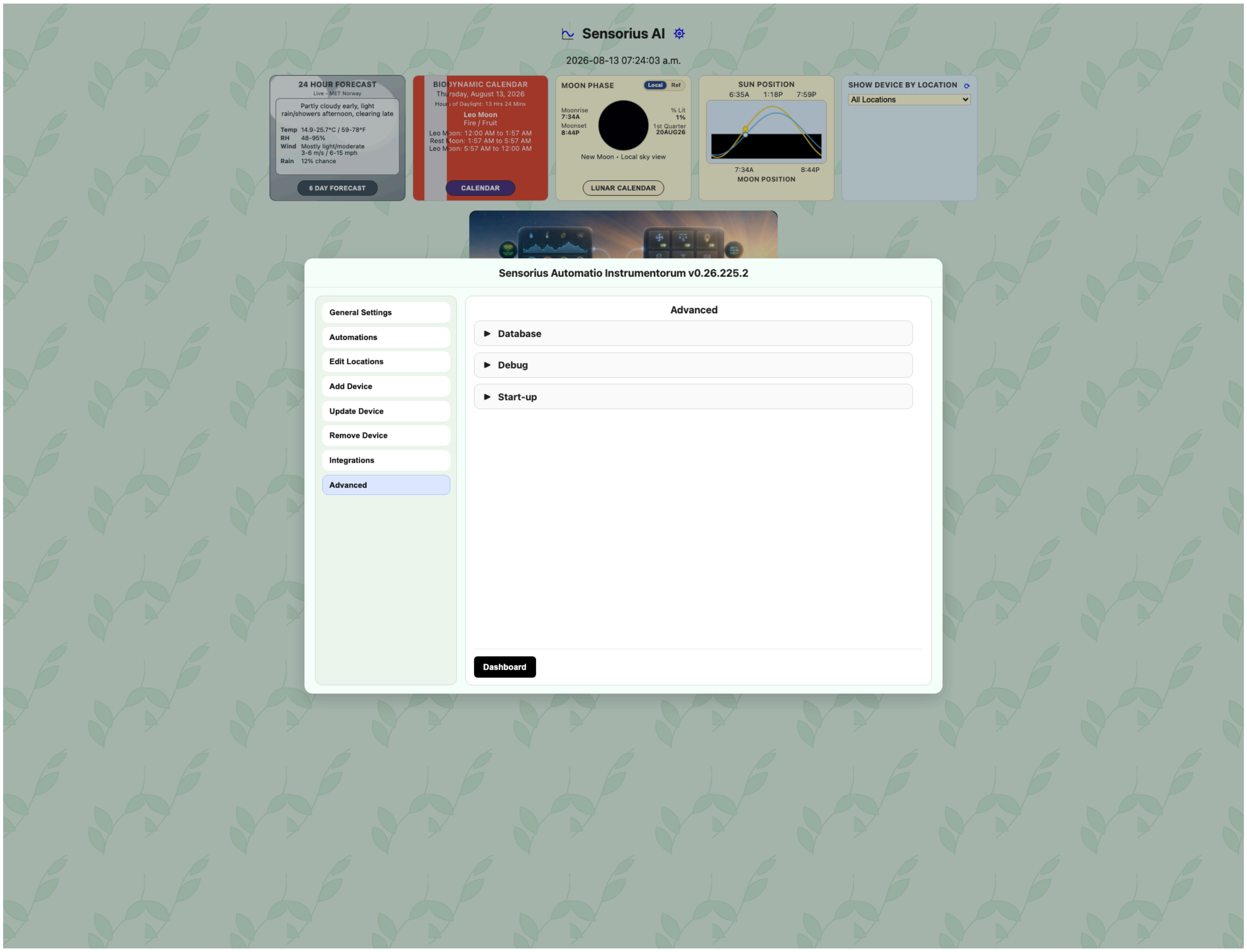
Fields and selectors:

- **MQTT Interface**: enables or disables WeeWX MQTT ingest. Options are **Disabled** and **Enabled**.
- **Runtime state**: shows whether Sensorius is receiving WeeWX data, the latest sample, age, and offline timing.
- **WeeWX Database**: read-only database path, normally `/var/lib/weewx/weewx.sdb` for a WeeWX host.
- **Sensorius Sensor ID**: sensor ID Sensorius uses for WeeWX readings. Default is `weewx-station`.
- **MQTT Topic Filter**: MQTT subscription filter for WeeWX messages. Default is `weewx/#`.
- **Update Period Seconds**: expected update period, from 15 to 3600 seconds.
- **Sensorius Broker**: read-only broker host and port used by Sensorius.
- **Save**: writes the `[WeeWX]` settings and creates WeeWX sensor settings if needed.

When WeeWX runs on the same host and `/etc/weewx/weewx.conf` or `/home/weewx/weewx.conf` is readable, Sensorius records the active station model in the WeeWX station sensor settings. The Sensor Settings **Sensor Info** pane shows it as `Station: <model>`.

If the MQTT topic changes, Sensorius applies the subscription update live when MQTT ingest is running. If MQTT ingest is not running, the saved setting applies when MQTT ingest starts.

Advanced Pane



Advanced settings affect startup, logging, and stored data. Change them only when you understand the impact.

Database

- **Maximum Days of Data (30-365)**: database retention window. Valid range is 30 to 365 days. This affects how much history graphs can show.
- **Archive Database**: creates a SQLite database snapshot under `database_archives/` next to the active database and downloads the snapshot.
- **Renew Database**: archives the current SQLite database, deletes the active database files, and creates a new empty database. This is an intentionally drastic recovery action.
- **Save**: writes only the Database section.

Debug

- **SENSORIUS_FILE_LOG**: enables file logging when checked.

- **SENSORIUS_LOG_LEVEL**: logging detail. Options are **DEBUG**, **INFO**, **WARNING**, **ERROR**, and **CRITICAL**.
- **SENSORIUS_DEBUG_MODULES**: module-specific debug checkboxes. Options are loaded from the app's advanced status endpoint.
- **Save**: writes only the Debug section.

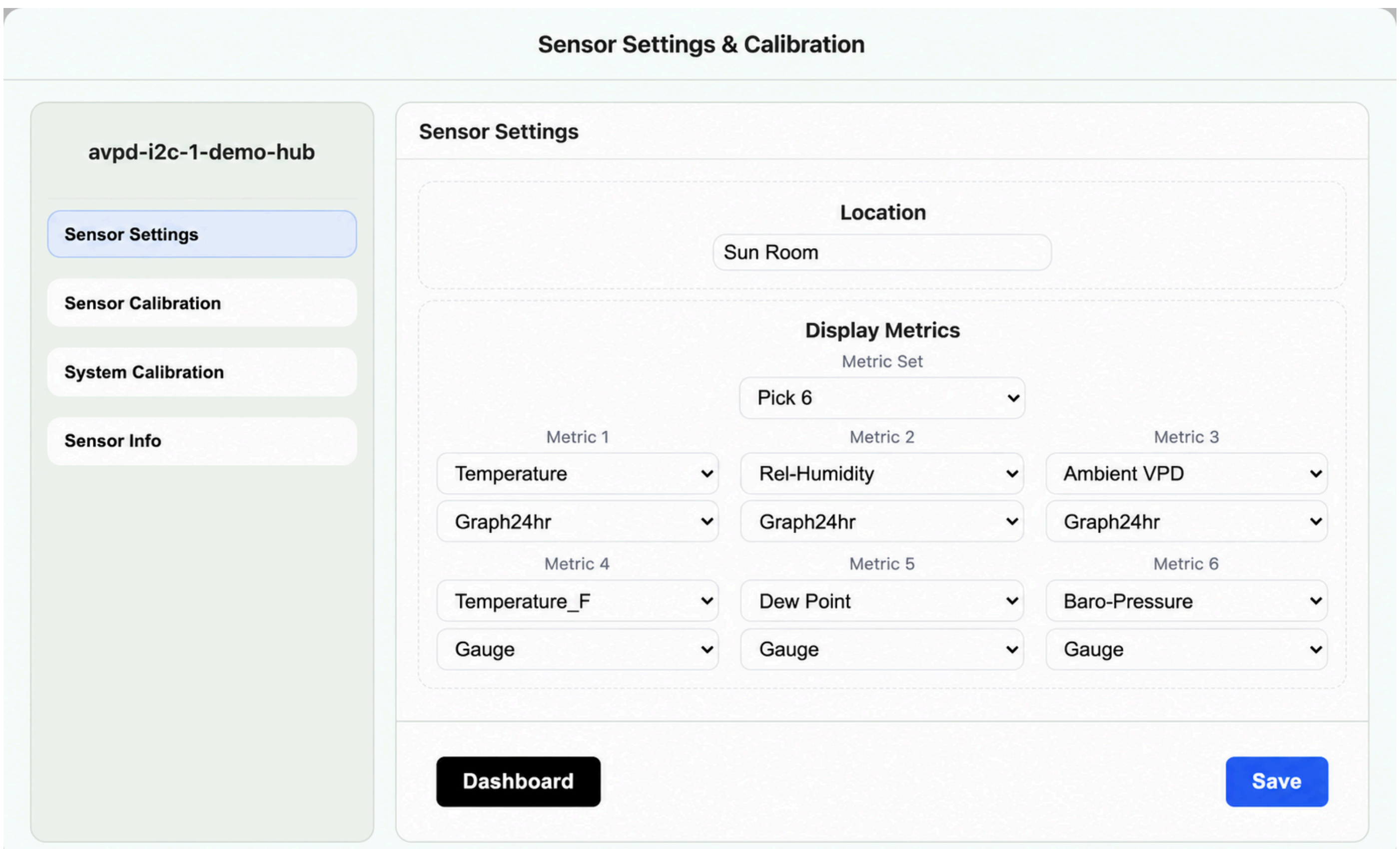
Start-up

- **Auto-start Sensorius on login**: creates or removes an auto-start entry for Sensorius.
- **Auto-start scope**: **User-level (default)** starts for the current user. **System-level** is for system service style installs and may require elevated permissions outside the web UI.
- **Save**: writes only the Start-up section.

Sensor Settings

Open Sensor Settings from a sensor card when you need to organize a sensor, choose which readings appear on the dashboard, calibrate readings, or check device health.

Sensor Settings Pane



Fields and selectors:

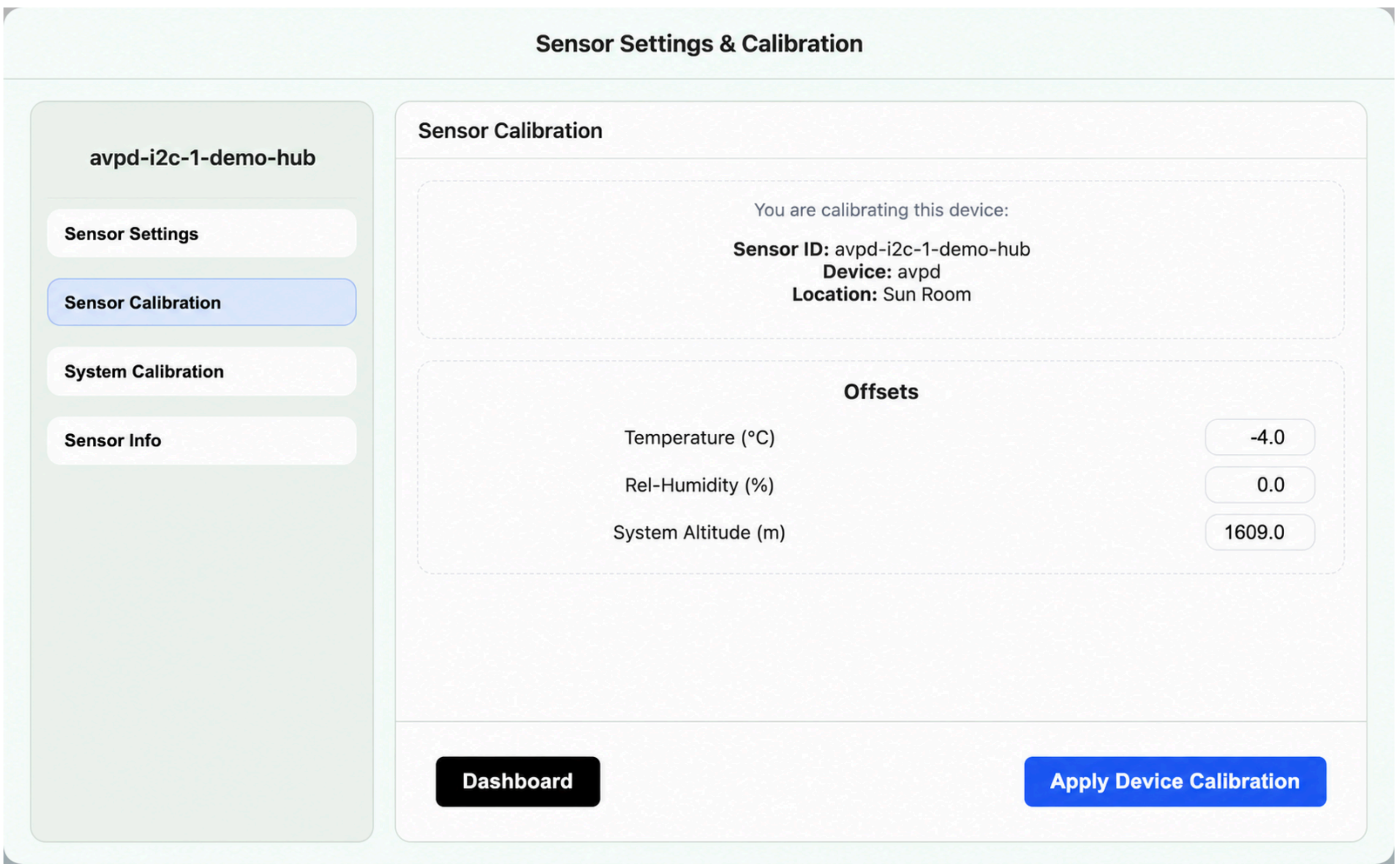
- **Location**: the practical place where the sensor is installed, such as Greenhouse, Seed Rack, Bed 2, Propagation Tent, or North Field. This is saved in that sensor's `sensor.toml` and is also used by the dashboard, location editor, and automation selector labels.

- **Metric 1-6**: dashboard metric slots. Options come from the sensor's available metric list. That list is built from device metadata, known database metrics, and Sensorius gauge configuration.
- **Display Style 1-6**: display style for each selected metric. Options are **Gauge**, **6Hr Graph**, and **24Hr Graph**.
- **Dashboard**: closes the modal and returns to the dashboard.
- **Restart Device**: appears for devices that support remote restart, such as supported Nodus devices. It sends a restart request to the device.
- **Save**: writes only the Sensor Settings pane values to `sensor_settings/<sensor_id>/sensor.toml`. Calibration and Sensor Info pane state are not submitted. For remote Nodus devices, Sensorius also pushes a correlated settings update over MQTT when needed.

Use clear location names. They are visible to nontechnical users and make later automation rules easier to understand.

Metric slot order is authoritative in **Pick 6** mode: Metric 1 is the leftmost tile and Metric 6 is the rightmost. Display Style 1 applies to Metric 1, Display Style 2 to Metric 2, and so on. Saving these fields is the supported way to make a dashboard order or display style persistent.

Sensor Calibration Pane



The Sensor Calibration pane adjusts the selected physical device. It is meant for cases where one sensor is consistently high or low compared with a trusted reference.

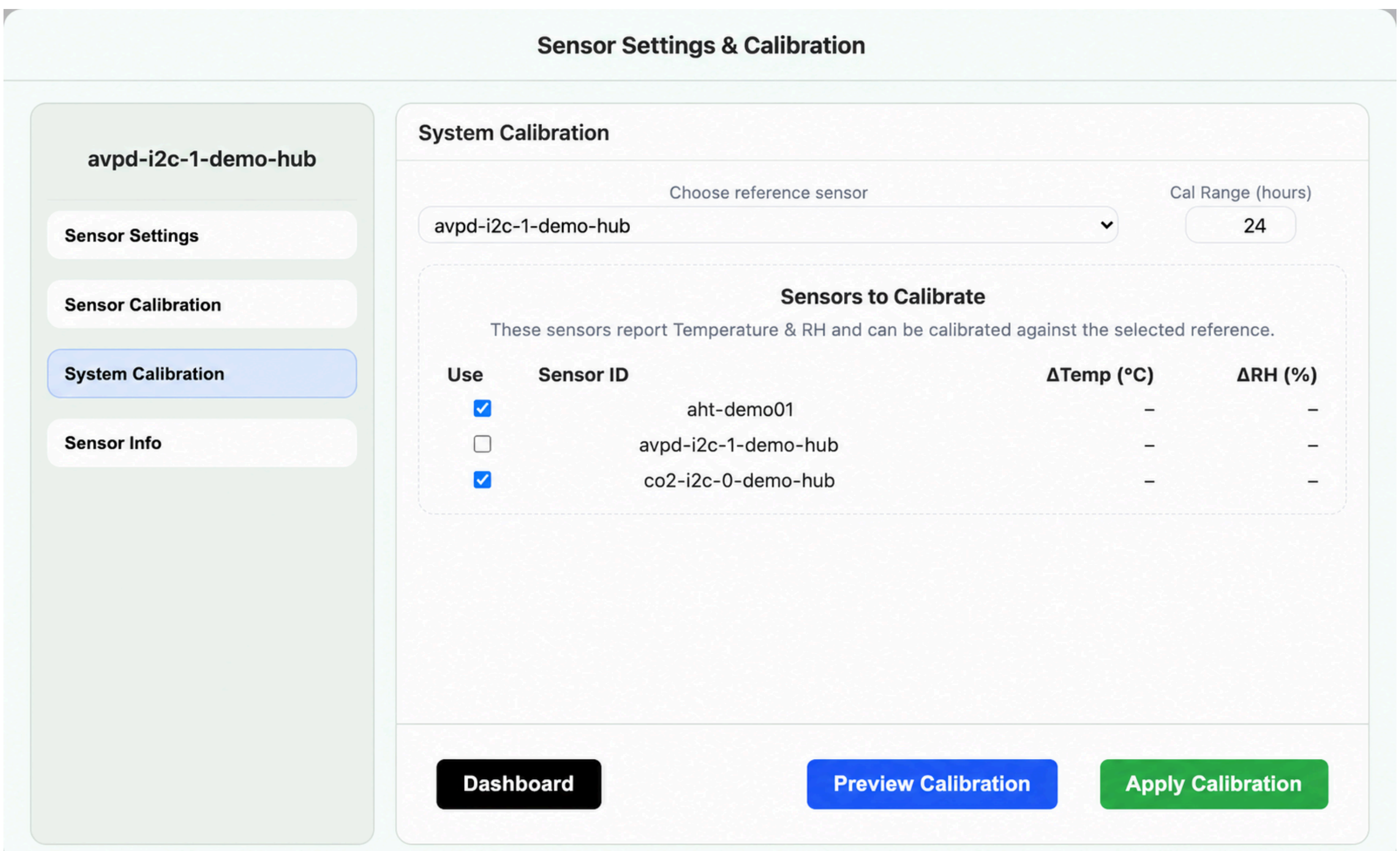
Common fields and controls:

- **Sensor ID**: the Sensorius identifier for this sensor.

- **Device**: device type or label from local settings or Nodus metadata.
- **Location**: the sensor location from Sensor Settings.
- **Ambient Temperature Offset (C)**: correction added to the ambient temperature channel on APVPD-style devices.
- **Ambient RH Offset (%)**: correction added to the ambient relative humidity channel on APVPD-style devices.
- **Altitude Calibration / Offsets**: device-specific numeric offsets exposed by the sensor. Available fields depend on the sensor type and firmware metadata.
- **Calibrate Plant Sensor**: runs the APVPD plant-sensor calibration routine when the device supports it.
- **Calibrate pH 4.0 / 7.0 / 10.0**: soil-sensor pH buffer calibration buttons. Use the matching buffer solution and wait for the probe to stabilize before applying.
- **Soil Offsets**: manual numeric offsets for soil sensor channels such as pH or other exposed calibration values.
- **Apply Device Calibration**: saves the device calibration. For Nodus devices, the command is sent to the device and the local settings shadow is updated.

Calibration data comes from the sensor's `Calibration` section, Nodus metadata, and device-specific calibration endpoints. Apply small changes, then watch the dashboard and graphs to confirm the readings now track your trusted reference.

System Calibration Pane



System Calibration compares multiple temperature/RH sensors to a reference sensor over recent history.

Fields and selectors:

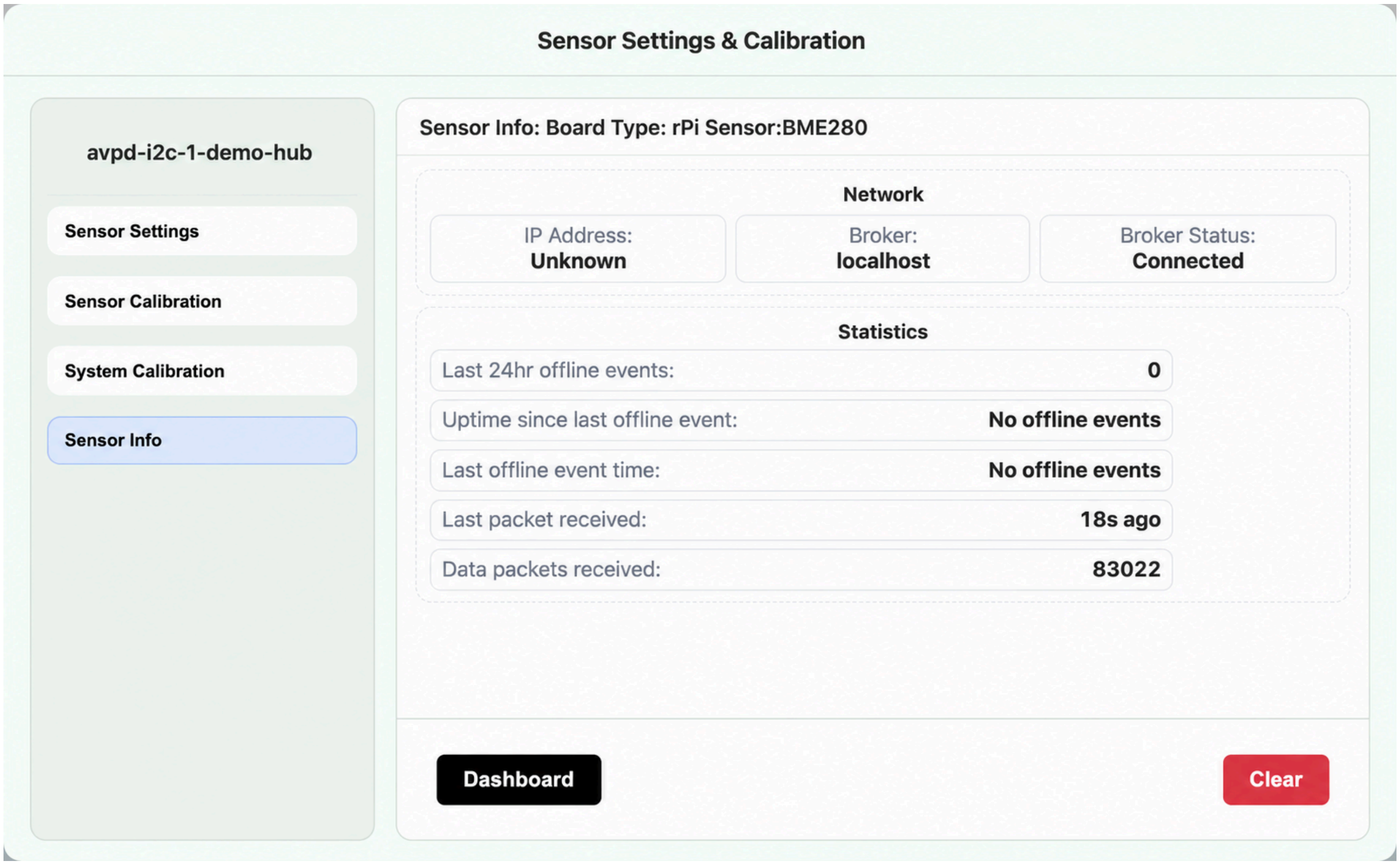
- **Choose reference sensor**: the trusted sensor used as the baseline. Options come from sensors that report temperature and relative humidity.
- **Cal Range (hours)**: the history window used for comparison. Options are numeric values from 24 to 72 hours.
- **Use**: checkbox for each candidate sensor. Checked sensors are included in the preview and apply step.
- **Sensor ID**: candidate sensor name.
- **Delta Temp (C)**: previewed temperature correction needed to align with the reference.
- **Delta RH (%)**: previewed relative-humidity correction needed to align with the reference.
- **Preview Calibration**: calculates corrections from stored readings without applying them.
- **Apply Calibration**: applies the previewed corrections. It is disabled until a valid preview is available.

This tool reads historical samples from the SQLite database. It works best when sensors have been near each other long enough to collect comparable data.

For the best calibration results:

1. Make sure the Nodus sensors have been added to Sensorius.
2. If possible, place a trusted temperature and relative-humidity gauge with the Nodus sensors. A gauge that records 24-hour minimum, average, and maximum values is especially useful.
3. Before deploying the Nodus sensors to their final locations, place them together in an area that experiences meaningful variation, ideally about 10 C of temperature change and 10 percentage points of relative-humidity change within 24 hours.
4. Allow Sensorius to collect temperature and relative-humidity data from the Nodus sensors for at least 24 hours and up to 72 hours.
5. After at least 24 hours, compare each Nodus sensor's 24-hour minimum, average, and maximum values with the trusted gauge. If you do not have a trusted gauge, compare the temperature and relative-humidity graphs and choose the Nodus sensor that appears to provide the best representation as the reference.
6. Select the reference sensor, check each additional sensor that you want to calibrate, and click **Preview Calibration**. Review the corrections, then click **Apply Calibration**.

Sensor Info Pane



The Sensor Info pane is a health and diagnostics view.

Fields shown:

- **IP Address**: the last known device IP address from Nodus metadata or runtime network information. Local sensors may show Unknown.
- **Broker**: MQTT broker the remote device is using, when known.
- **Broker Status**: remote broker connection status, when reported.
- **Last 24hr offline events**: offline transitions recorded during the last 24 hours.
- **Uptime since last offline event**: time since the latest offline event, or since you cleared local counters.
- **Last offline event time**: timestamp of the last offline event.
- **Last packet received**: age or timestamp of the most recent packet from the sensor.
- **Data packets received**: packet count known to Sensorius.
- **Clear**: clears the browser-side display baseline for these counters. It does not delete historical database rows.

Use this pane when a sensor appears stale, missing, or unstable.

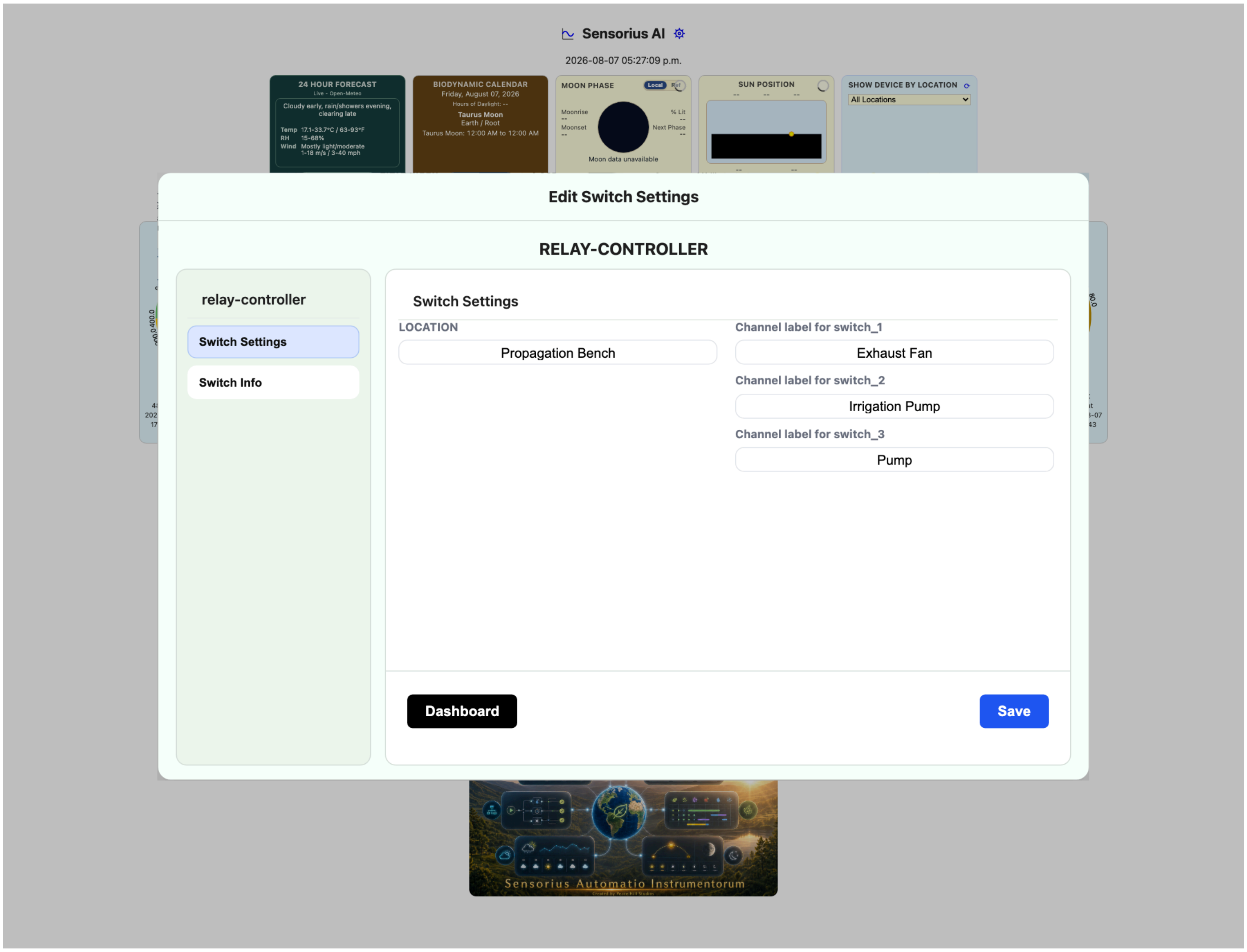
Switch Settings

Switches control things such as pumps, fans, lights, valves, heaters, vents, and other relay-driven equipment. A switch may be a local Raspberry Pi relay board or a remote Nodus switch. Sensorius presents both through the same dashboard and settings screens.

Switch state changes are stored as switch events. These events are available for dashboard state, recent-event lists, full-screen graph overlays, and automation review.

Open Switch Settings from a switch card when you need to label channels, set the switch location, or check switch health. Create and edit rules separately under **General Settings > Automations**.

Switch Settings Pane



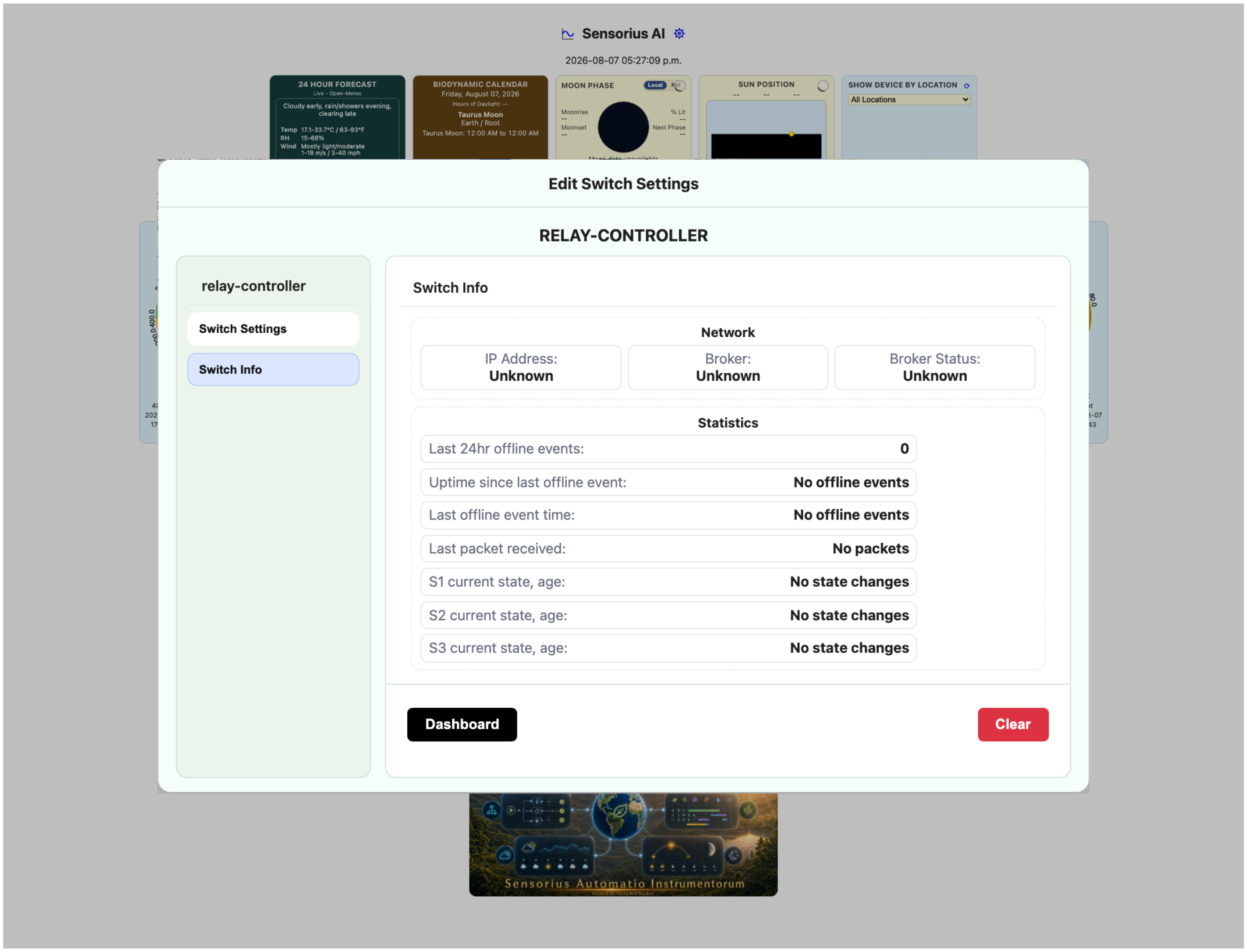
Fields and controls:

- **Location**: where the switch device or relay box is installed. This is saved in `switch_settings/<switch_id>/switch.toml` and is shown on the dashboard and location editor.
- **Channel label for switch_N**: friendly label for each channel, such as Exhaust Fan, Irrigation Pump, Heat Mat, North Vent, or Lights. Labels are saved in switch settings.
- **Dashboard timer gear**: opens the channel's manual auto-off timer. Use `0` to disable it or 30-9999 seconds; the dashboard rounds entries to a 30-second step. The value lasts only until Sensorius restarts.
- **Dashboard**: closes the modal.
- **Restart Device**: appears for supported remote switches and sends a restart request.

- **Save**: writes only the Switch Settings pane values. Switch Info is not submitted. For remote Nodus switches, Sensorius also sends a settings update over MQTT when needed.

Keep labels stable for operator clarity, but the internal switch address is the stable `<switch_id>::<channel_id>` key.

Switch Info Pane



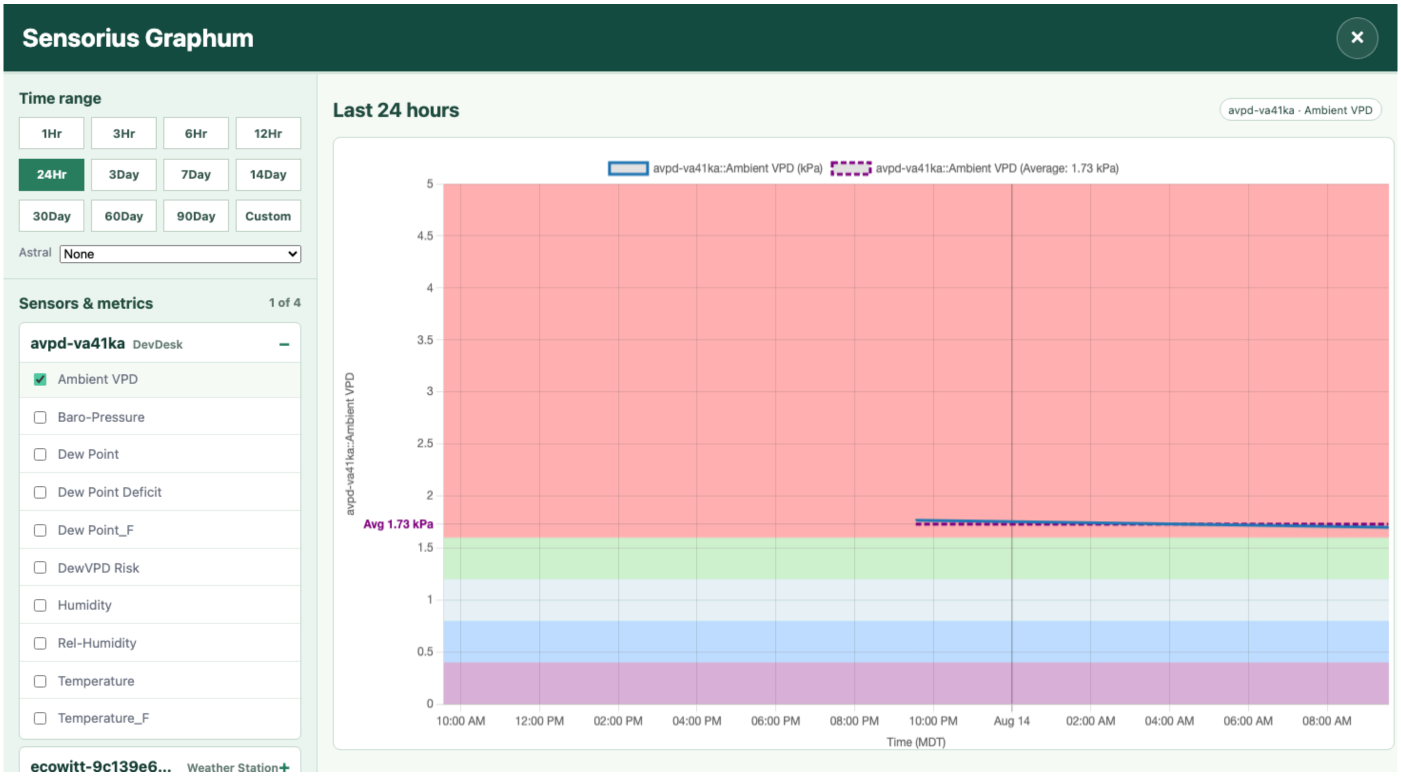
The Switch Info pane shows:

- **IP Address**: last known IP address for a remote switch.
- **Broker**: MQTT broker reported by the switch.
- **Broker Status**: remote broker connection status.
- **Last 24hr offline events**: offline transitions during the last 24 hours.
- **Uptime since last offline event**: time since the latest offline event or local clear baseline.
- **Last offline event time**: timestamp of the last offline event.
- **Last packet received**: age or timestamp of the most recent packet.
- **Switch current state, age**: per-channel state and how long that state has been known.
- **Clear**: clears the browser-side display baseline for statistics.

Use this pane when commands do not seem to reach a switch or when a remote relay appears to drop offline.

Sensorius Graphum

Open **Sensorius Graphum** when you want to compare readings over time, investigate spikes, or see whether a switch action changed the environment.



The full-screen graph displays up to four selected observations. The first selected metric uses the left axis, and additional metrics use the right axis. A switch channel counts as an observation and appears as ON/OFF transition markers. When average data is available, a purple dashed line labeled **Average** includes the arithmetic-average value for the selected visible window. The same value appears as a bold purple **Avg** tick at the line's height on its value axis. VPD graphs show VPD range coloring, and some metrics show gauge-zone background bands. These colored bands come from metric display zones, not automation thresholds.

On the dashboard, a WeeWX **Wind Direction** card configured as a **6Hr Graph** or **24Hr Graph** displays a wind rose instead of a direction line. Sixteen direction sectors show how frequently wind arrived from each compass direction. Stacked blue bands show wind speed in **0-5**, **5-15**, **15-30**, and **30+ mph** ranges, progressing from light blue for low speed to dark blue for high speed. The card title changes to **Wind-Rose (6hr)** or **Wind-Rose (24hr)** for the selected graph window and returns to **Wind Direction (deg)** in compass mode.

Each wind rose uses only observations from its selected 6-hour or 24-hour window. Sector lengths are normalized independently: the most frequent direction in that window reaches the outer ring, and the other sectors are scaled relative to it. Therefore, sector lengths should be compared within one rose, not as an absolute scale between the 6-hour and 24-hour roses. The center label gives the dominant compass direction and its actual percentage of samples for the selected window.

The large **mph** value below the rose is the latest wind-speed reading and may change as live station data arrives. The **Min**, **Avg**, and **Max** row is intentionally fixed to the trailing 24 hours in both wind-rose views; it does not change to a 6-hour summary when the 6-hour rose is selected.

Switch event overlays appear as vertical lines. The legend shows which colors mean ON and OFF for each selected switch channel.

Sensorius Graphum Controls

The graph button opens the full-screen workspace directly. Its scrollable left pane contains:

- **Time range**: select **1Hr**, **3Hr**, **6Hr**, **12Hr**, **24Hr**, **3Day**, **7Day**, **14Day**, **30Day**, **60Day**, or **90Day**. Available longer ranges depend on the configured database-retention period. Select **Custom** to enter an exact start and end date and time.
- **Astral**: select **None**, **Sun**, **Moon**, or **Sun & Moon** to control the optional Astral position panel shown with the graph.
- **Sensors & metrics**: select the **+** beside a sensor to expand it, then check one or more metrics. Each device location appears to the right of its stable device ID. Select **–** to collapse the sensor without changing its checked metrics.
- **Switches**: expand a switch and check a channel to overlay its ON/OFF transitions on the selected metric history. The switch location appears to the right of its stable device ID. A switch overlay accompanies at least one sensor metric; it does not form a graph by itself.
- **Selection count**: sensor metrics and switch channels share a maximum of four selections. At **4 / 4**, unchecked boxes are disabled. Clear any checked item to free a slot.

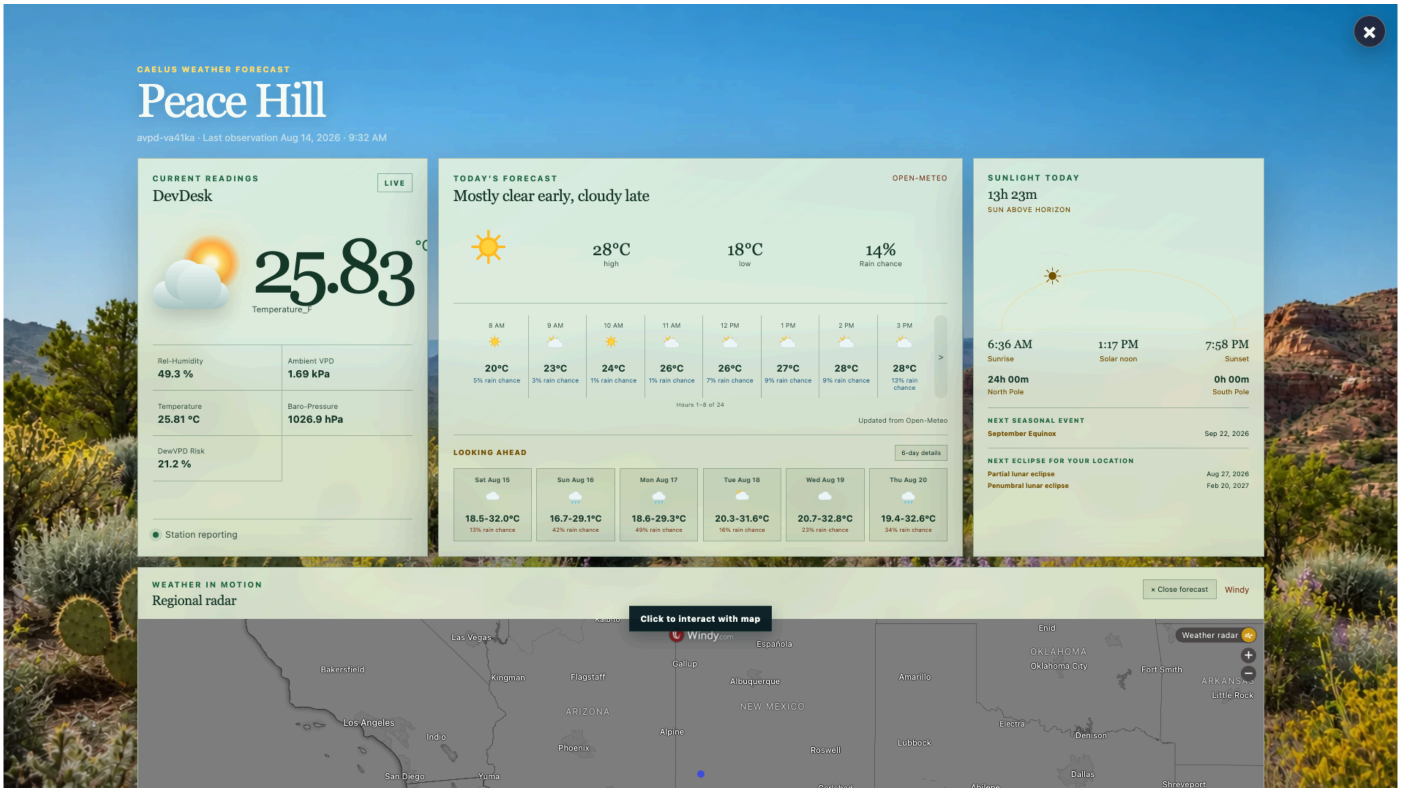
Checked items also appear as labeled chips above the chart, so the active graph can be confirmed without reopening a collapsed group. Every selection or time change redraws the graph automatically; there is no separate **Graph It** step. The first selected metric uses the left value axis and subsequent metrics use separate right-side axes. Switch channels appear as colored transition markers.

While the workspace is open, new sensor readings trigger a refresh through the dashboard's live-update path, switch events trigger an immediate refresh, and a 15-second fallback refresh keeps the graph current if a live notification is missed. Hidden or closed graphs do not poll, and a newer refresh cancels any older graph request still in progress. Graph selections are temporary and are not saved as named graph sets. Select **×** in the upper-right to close the workspace and return to the dashboard.

Use switch overlays to answer practical questions: whether a fan cooled the greenhouse, whether irrigation raised soil moisture, or whether lights changed VPD.

Caelus Weather Forecast

Select **6 Day Forecast** on the dashboard forecast card to open the integrated Caelus full-screen weather display at `http://<sensorius-host>:8000/weather-forecast`. Select the circled **×** in the upper-right to close the forecast and return to the Sensorius dashboard.



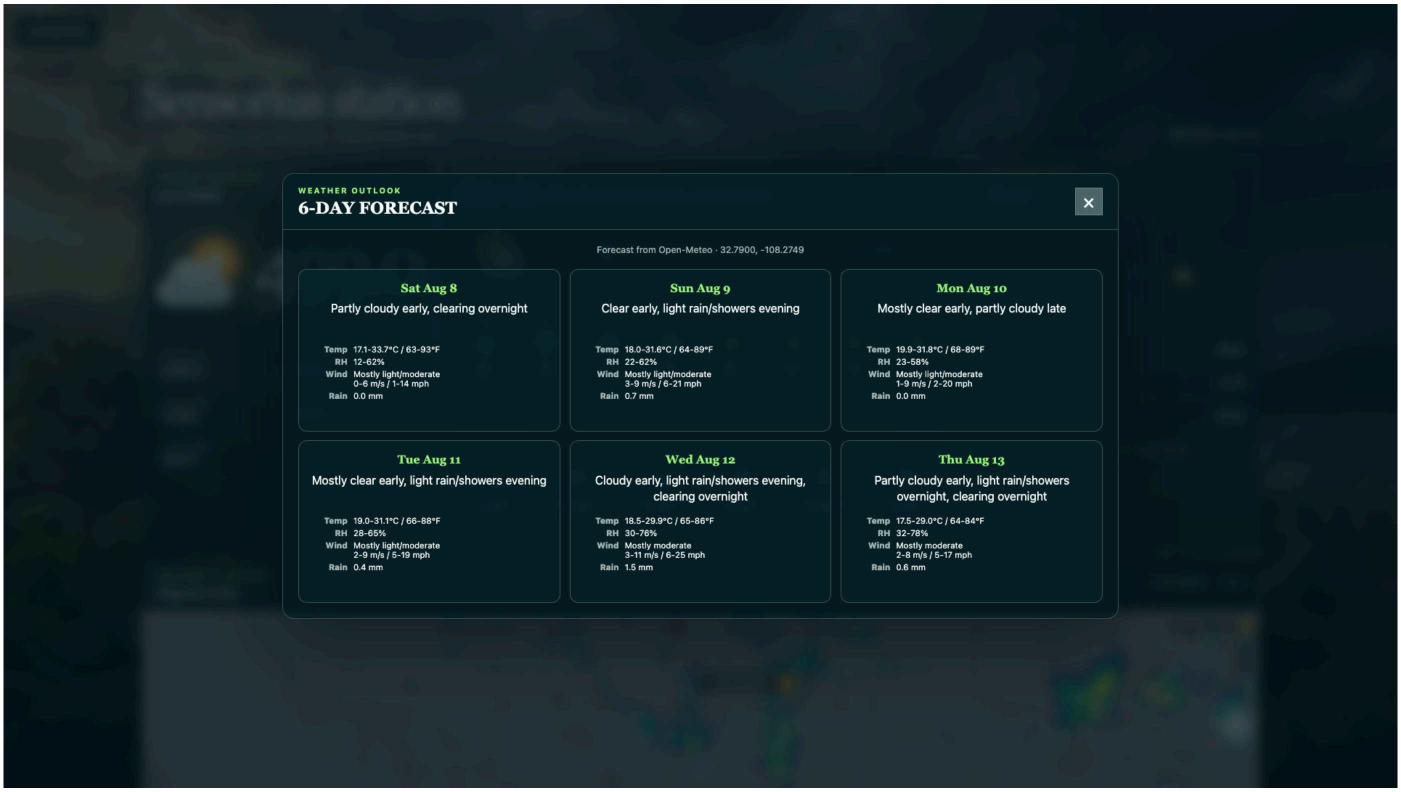
The weather display uses the existing Sensorius Astral latitude, longitude, timezone, and optional Community/Location Name. Its top row presents the selected sensor's latest current readings, the canonical Sensorius forecast, and the sunrise/sunset daylight track. Hourly forecast times use the local 12-hour AM/PM clock. The sunlight card also shows current North and South Pole daylight, the next seasonal event, and up to three solar or lunar eclipses visible from the configured Astral location during the next twelve months. A full-width Windy map opens in radar view below that row, followed by the current Moon and phase cycle at the bottom. The display also provides theme-matched six-day details. It does not run a separate weather service or maintain a separate settings file or readings database.

The Windy map is initially interaction-locked so the mouse wheel continues to scroll the Caelus page. Select **Click to interact with map** on the map's top border to enable its zoom and pan controls; move the pointer outside the map (or press Escape) to lock it again.

The lunar panel keeps the live Moon and its current illumination, lunar age, altitude, and local orientation in the center. The four most recent phase milestones appear chronologically on the left, and the next four appear on the right, with each phase's local date below its name. Every disk is oriented for the configured Astral location near the Moon's highest point on that date. Full-moon milestones use familiar traditional names such as **Harvest Moon**.

Current Readings displays the selected sensor's configured **Display Metrics** in their saved order, including their standard Sensorius units, and identifies the source by its configured sensor **Location**. This allows a weather station, a Nodus environmental sensor, or a WeeWX station with fewer metrics to use the same full-screen view without requiring weather-only metric names. While Caelus remains open, Current Readings follows the selected sensor's collection interval (normally 60 seconds for Nodus and the configured polling interval, 300 seconds by default, for Ecowitt). Today's Forecast refreshes at the top of each hour, while Sunlight Today continues to refresh every 300 seconds.

The hourly strip presents the next 24 hours as one-hour forecasts, with eight hours visible at a time. Use the narrow left and right controls to move the window one hour; each control disappears when its edge is reached. Hourly and daily forecasts show rain or snow chance percentages, while condition icons identify rain or snow when the provider predicts a precipitation type. Select **6-day details** under **Looking Ahead** to open the detailed outlook with daily conditions, temperature, relative humidity, wind, and rain or snow chance.



Biodynamic Calendar

Sensorius includes the full Biodynamic Calendar application. It runs in the Sensorius web process and opens as a full-screen page when the dashboard **Calendar** button is selected.

The built-in month view includes a color legend for Root, Leaf, Flower, Fruit, Rest, and Transition periods above the calendar grid.

The calendar uses Sensorius Astral settings for latitude, longitude, altitude, and timezone. Notes, summaries, planting records, and computed cache entries are stored in the Sensorius SQLite database. It is available at `http://<sensorius-host>:8000/calendar`; select the circled **×** in the upper-right to close the calendar and return to the dashboard.

Full-Screen Calendar

The full-screen calendar shows:

- Current biodynamic sign, element, and plant focus.
- Current day's biodynamic windows.
- Month navigation.
- Day cells colored by dominant biodynamic influence.

- Daily Summary for the selected day.
- Daily Notes for the selected day.
- Save Note.
- Print Report for the selected month calendar with dated biodynamic hints, cached daily summaries, and notes.

Printing Calendar Reports

On the Raspberry Pi desktop shell, **Report** uses the Pi's default CUPS printer. Raspberry Pi deployment offers to configure a permanent driverless network-printer queue. If the printer was connected later, an administrator can rerun `/home/<user>/Sensorius/scripts/setup_rpi_printer.sh` once; ordinary report users do not need to configure the printer for every report.

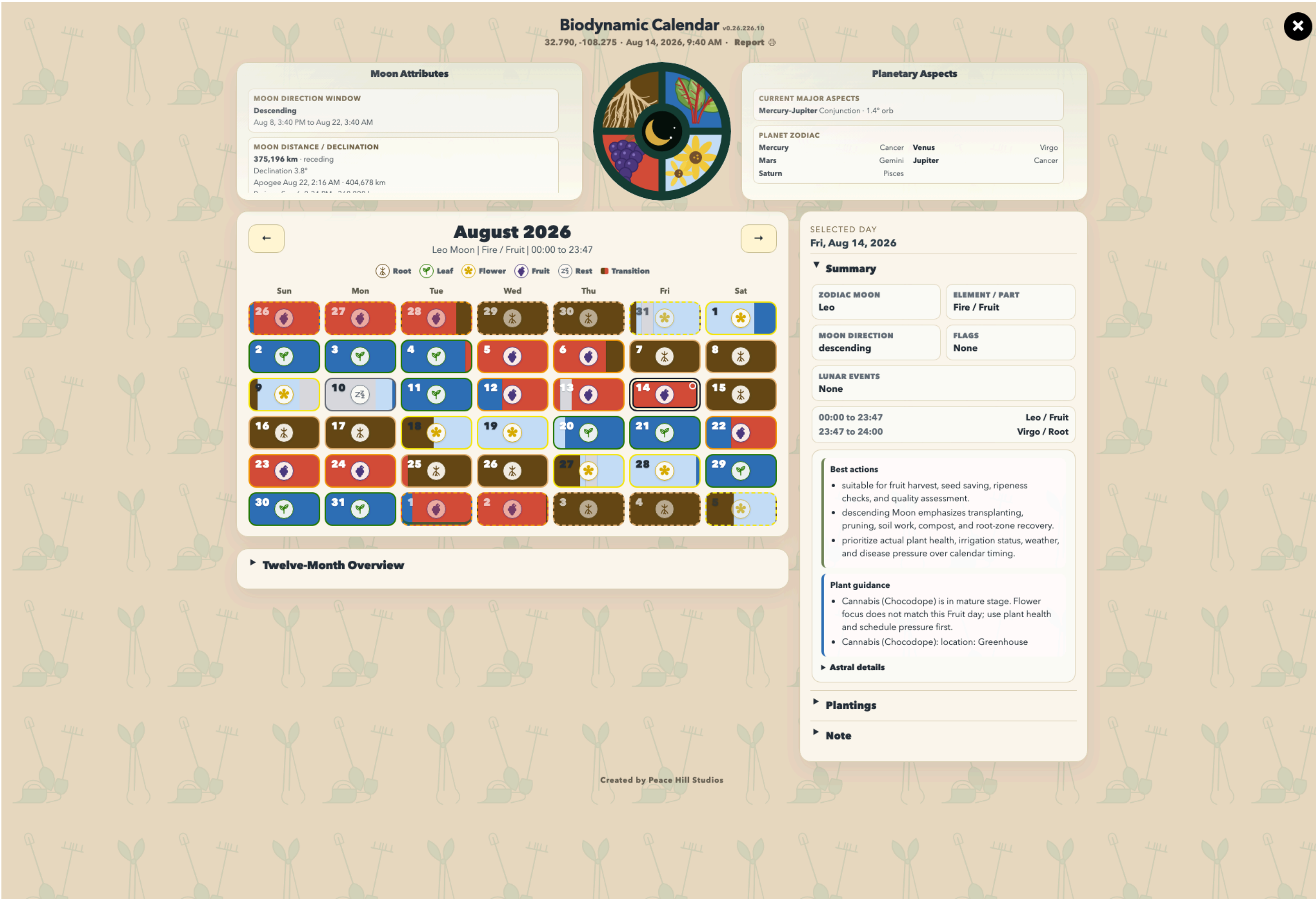
When the calendar is opened from another computer or mobile device, printing uses that device's browser and printers instead of the Raspberry Pi printer. Saving as PDF also occurs on the device displaying the print dialog.

Before relying on local Pi printing, confirm that its print dialog shows one enabled default destination. If duplicate entries or a disabled destination appear, use the printer troubleshooting section in the Operations guide.

Daily summaries come from Sensorius' biodynamic summary storage and are generated on demand when a day is selected. Browsing future months does not generate summaries for their default day. For the current day, the summary may include a **24hr Forecast** section if weather forecast data is enabled in General Settings.

The dashboard Biodynamic Calendar card remains available for a quick current-status view. The Calendar button opens the full application for month planning, planting records, notes, daily guidance, and reports.

Integrated Biodynamic Calendar Features



The integrated application provides:

- A full-screen calendar UI.
- Moon Attributes panel with lunar direction, distance, declination, and eclipse information.
- Planetary Aspects panel with current major aspects and planet zodiac placements from Skyfield.
- Sensorius-managed Astral location and timezone.
- Twelve-Month Overview.
- Planting records with crop details.
- Notes and print reports.
- Sensorius SQLite storage for notes, summaries, planting records, and cache entries.

Companion app fields and controls:

- **Location**: comes from the Astral and Time sections in Sensorius General Settings. Change it there to invalidate and rebuild calendar data.
- **Previous / Next month arrows**: move the main month calendar.
- **Calendar day cells**: select a day. The selected day drives the Daily Summary, selected facts, notes, and planting context.
- **Twelve-Month Overview**: shows a longer planning range assembled from the shared background cache.
- **Daily Summary**: explains the selected day, including biodynamic focus and relevant timing.

- **Plant**: crop or plant name, such as Tomato.
- **Variety**: cultivar or variety name.
- **Focus**: biodynamic plant focus. Options are Auto, Root, Leaf, Flower, and Fruit. Auto lets the app infer the focus from crop information when possible.
- **Start**: start method. Options are Seed and Transplant.
- **Start Date**: planned or actual seeding/transplant date.
- **Harvest**: expected harvest date.
- **Days to Maturity**: optional number from 1 to 730 days.
- **Location**: bed, greenhouse, field, tray, or other practical planting location.
- **Plant Type**: descriptive crop class, such as fruiting vegetable.
- **Attributes**: free-form notes for spacing, succession, hardening, trellis, harvest window, or other crop details.
- **Save Planting**: stores or updates the planting record.
- **Clear**: clears the planting form.
- **Edit / Delete** in the planting list: updates or removes an existing planting record.
- **Your Notes**: free-form note for the selected day.
- **Save Note**: stores the note for that date.
- **Print**: prints the selected calendar/report view.

No separate Biodynamic Calendar service, port, database path environment variable, or startup entry is required.

Good Operating Habits

- Use names and locations that match how people talk about the garden or farm.
- Test automations with harmless loads before connecting pumps, heaters, valves, or lights.
- Use hysteresis and short delays on threshold rules.
- Watch graphs after calibration changes.
- Keep MQTT broker addresses stable once Nodus devices or Home Assistant depend on them.
- Back up `sensor_settings/`, `switch_settings/`, `system_settings/`, and `sensorius_data.db` before major changes.
- Restart Sensorius after changes that affect startup, host binding, service setup, or MQTT subscriptions when the UI or save message indicates a restart is needed.

Opening Sensorius

If you installed Sensorius as a service, it starts automatically when the host computer starts or restarts. Wait for the host computer or Raspberry Pi to finish starting; the dashboard will appear after a few seconds. Service installs continue running in the background. To restart a Linux or Raspberry Pi systemd service manually, open a terminal and enter:

```
sudo systemctl restart sensorius
```

If you start Sensorius manually, run this from the installed Sensorius folder (the default is `~/Sensorius`):

```
cd ~/Sensorius
.venv/bin/python Sensorius.py
```

The application will start and display the Sensorius dashboard. The first run may take a little longer while it sets up the system's location, database, and calendar data.

The dashboard can also be opened using one of these addresses:

- Same computer: `http://127.0.0.1:8000`
- Local network hostname: `http://<hostname>.local:8000`
- Another device on the same network: `http://<sensorius-host-ip>:8000`

Keep the terminal process running for manual starts.

Keep Sensorius On A Trusted Network

Sensorius is built for a trusted private LAN. It does not provide a complete login/session boundary around every settings, onboarding, calibration, switch, or maintenance action. The optional web API key protects selected operations, not the whole UI.

Do not expose the Sensorius HTTP or MQTT ports directly to the Internet or use router port forwarding. Keep the hub and devices behind a firewall, use an isolated IoT VLAN where appropriate, and use a VPN or a trusted remote-access application such as RustDesk for remote access. On a Raspberry Pi, you can use either RustDesk or Raspberry Pi Connect. If only the host computer needs the UI, set `SENSORIUS_HTTP_HOST=127.0.0.1` in `.env` and restart Sensorius. Treat `.env`, runtime settings, backups, and diagnostic exports as sensitive because stored secrets are not protected by strong encryption.

Where Sensorius Gets Its Information

Live sensor readings come from local Raspberry Pi sensor controllers and from MQTT-discovered Nodus devices. Sensor readings are written to the local SQLite database, `sensorius_data.db`, in the Sensorius process working directory unless the service or caller passes a different database path. Many service installs use the runtime directory, such as `/Users/<user>/Sensorius/sensorius_data.db` on macOS or `/home/<user>/Sensorius/sensorius_data.db` on Linux.

Sensor and switch names, locations, display choices, calibration offsets, and channel labels come from Sensorius settings files under the macOS runtime settings folder:

- System settings: `/Users/<user>/Sensorius/system_settings/<device_id>/settings.toml`
- Sensor settings: `/Users/<user>/Sensorius/sensor_settings/<sensor_id>/sensor.toml`
- Switch settings: `/Users/<user>/Sensorius/switch_settings/<switch_id>/switch.toml`
- Advanced automations: `/Users/<user>/Sensorius/switch_settings/automations/automations.toml`

For Nodus devices, Sensorius also listens for MQTT metadata and state messages. That metadata tells Sensorius what a device is, which readings or channels it provides, whether it is online, and which remote settings can be updated.

Troubleshooting

If a sensor does not appear:

- Confirm Sensorius is running and the dashboard is reachable.
- For Nodus devices, confirm the device is powered, on the expected Wi-Fi, and publishing MQTT metadata.
- Check **Sensor Info** for last packet and offline events.
- Refresh the dashboard after onboarding or metadata changes.

If readings look wrong:

- Confirm the sensor is in the intended location.
- Compare against a trusted reference before calibrating.
- Use **Sensor Calibration** for one device and **System Calibration** for groups of similar temperature/RH sensors.
- Check whether the metric shown is the one you intended in **Sensor Settings > Metric 1-6**.

If switch control does not work:

- Confirm the switch is online in **Switch Info**.
- Check channel labels in **Switch Settings**.
- For Nodus switches, confirm MQTT command and state topics are flowing.
- Check whether an enabled Advanced automation is controlling the same channel.

If graphs have gaps:

- Confirm the sensor was online during the missing period.
- Check whether Sensorius or the host restarted.
- Check database retention in **General Settings > Advanced**.
- Confirm the selected metric exists for the selected sensor.

If Home Assistant does not show entities:

- Confirm Home Assistant integration is enabled.
- Verify broker host, port, TLS, username, and password.
- Confirm Home Assistant's MQTT integration is running.
- Restart Home Assistant or reload MQTT entities if retained discovery was recently changed.

If the Biodynamic Calendar is unavailable:

- Confirm latitude and longitude in **General Settings > Astral**, and confirm the timezone in **General Settings > Network Settings**.
- Confirm the normal Sensorius `/healthz` endpoint responds and that Sensorius can write to its SQLite database.
- If the calendar is still warming, leave Sensorius running for several minutes and verify the Astral location is complete.

Related Documentation

- Setup: `docs/setup.md`
- Operations: `docs/operations.md`
- Architecture: `docs/architecture.md`
- Configuration: `docs/configuration.md`
- Sensors and metrics: `docs/sensors.md`
- Switch automations: `docs/automations.md`
- MQTT: `docs/mqtt.md`
- Home Assistant: `docs/homeassistant.md`
- farmOS: `docs/farmos.md`
- Hardware: `docs/hardware.md`
- Integrated Biodynamic Calendar and companion migration: `docs/biodynamic_calendar_companion.md`
- Security policy and deployment boundary: `SECURITY.md`
- Third-party and binary notices: `THIRD_PARTY_NOTICES.md`